Choice-Free Reverse Mathematics.

From Barrow's Geometry to the Recursive Comprehension Axiom RCA_0

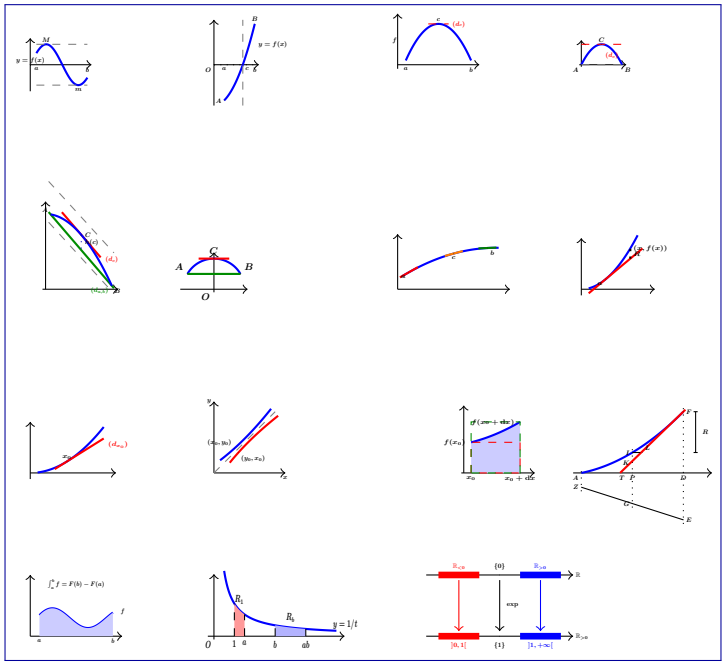
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Abstract

Part 1 develops classical single-variable real analysis in Barrow's geometric language of areas, ordinates, and subtangents. Part 2 classifies each theorem in reverse mathematics. MVT and Rolle are RCA_0 via Hardin–Velleman; the mean-value theorem for integrals, the inverse-function theorem (Yokoyama's strong-C^1 form), and the ln/exp chain are RCA_0; IVT, EVT, Darboux, and the weighted MVT are WKL_0 as proved here. All C^k hypotheses in Part 2 are read as strong C^k-coding; strength tags are upper bounds of the displayed proofs. No Part 2 entry uses Cauchy completeness or any form of Choice.



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Contents

1 A Geometric Development of Classical Real Analysis . . . 1
1. Axioms, definitions and geometric conventions . . . 1
2. Statements of the classical theorems . . . 3
2.1. Topological theorems . . . 3
2.2. Differentiation theorems . . . 3
2.3. Integration theorems . . . 5
2.4. The logarithm and the exponential . . . 5
3. Geometrical proofs . . . 6
3.1. Topological theorems . . . 6
3.2. Integration theorems . . . 12
3.3. The logarithm and the exponential . . . 16
4. Remarks on the geometric development . . . 17
2 Reverse-Mathematical Analysis in RCA_0 . . . 17
5. Coding conventions . . . 17
6. Research context . . . 18
6.1. The core hierarchy . . . 18
6.2. Illustrative theorems . . . 18
6.3. Dedekind vs. Cauchy completeness . . . 18
6.4. Constructive RM . . . 19
6.5. Lean 4 and Mathlib . . . 19
7. Axioms . . . 19
8. Postulates . . . 20
8.1. Integral postulates . . . 20
8.2. Differential postulates . . . 20
8.3. Logarithmic postulates . . . 20
8.4. Logical postulate . . . 21
8.5. External results . . . 21
9. Definitions . . . 21
10. Statements . . . 22
10.1. Topological theorems . . . 22
10.2. Differentiation theorems . . . 22
10.3. Integration theorems . . . 23
10.4. The logarithm and the exponential . . . 24
10.5. Illustrative theorems . . . 24
11. Proofs . . . 24
12. Scripts . . . 31
12.1. Foundational leaves . . . 32
12.2. Per-result trees . . . 32
12.3. Transitive closure . . . 33
12.4. Strength map . . . 33
12.5. Dependency depth . . . 34
12.6. Strength propagation . . . 34
13. Conclusion . . . 35
13.1. Wider hierarchy . . . 35
13.2. Dependency analysis . . . 35
13.3. Formal verification . . . 35

Introduction

Barrow's *Geometrical Lectures* (1670) give the first calculus textbook and the first geometric proof of the Fundamental Theorem of Calculus; see [1, 2].

Part 1 develops the classical theorems of single-variable real analysis — EVT, IVT, Fermat, Rolle, MVT and its refinements, convexity, inverse function, the Fundamental Theorems of Calculus, and the ln/exp addition and image theorems — in Barrow's geometric idiom, with figures and geometric proofs.

Part 2 classifies each theorem in reverse mathematics. Every axiom, postulate, definition and statement of Part 1 reappears as a tagged entry with a dependency table and an exact strength. Classification: all entries are RCA_0 except IVT, EVT, Darboux, and the weighted MVT, which are WKL_0. No Part 2 entry uses Cauchy completeness or any form of Choice [7, 14]. Coding conventions are in §5.

Part 1

A Geometric Development of Classical Real Analysis

Following Barrow [1, Lecture X, §11], we state and prove the classical theorems in Barrow's geometric idiom. See [2] for commentary.

1. Axioms, definitions and geometric conventions

Let $[a,\,b] \subseteq \mathbb{R}$ and let $f\colon [a,\,b] \to \mathbb{R}$ be continuous. The *area function of* f is the function $A_{a,f}$ defined for $x \in [a,\,b]$ by

$$A^{x}_{a,f} := \text{(signed) area of } \{(t,\,y) : a \le t \le x,\ \ 0 \le y \le f(t)\}, \tag{1}$$

and we write A^{x}_{a} for $A^{x}_{a,f}$ whenever the ambient function f is clear from context.

with Barrow's orientation: for all $u,\,w \in [a,\,b]$,

$$A^{u}_{u} = 0, \qquad A^{w}_{u} = -A^{u}_{w}, \tag{2}$$

and in particular $A^{x+\mathrm{d}x}_{x} = -A^{x}_{x+\mathrm{d}x}$ for $\mathrm{d}x < 0$. Equivalently, A is additive over intervals: $A^{w}_{u} = A^{v}_{u} + A^{w}_{v}$ for all $u,\,v,\,w \in [a,\,b]$, which is Equation 5. The curve $u \mapsto A^{u}_{a}$ is Barrow's AIF; $f(x_0)$ is his DE; the unit is his R.

Differentiability. f is differentiable at x if

$$\lim_{h\to 0}(f(x+h)-f(x))/h = f'(x) \tag{3}$$

exists.

Tangency (little-*o* form). Let G be defined on an open interval containing $x_0 \in \mathbb{R}$, and let $L(u) = m(u-x_0)+b$ be a line ($m,\ b \in \mathbb{R}$). Then G is differentiable at x_0 with tangent line L if and only if

$$L(x_0) = G(x_0) \quad \text{and} \quad G(x_0+\Delta x) = G(x_0) + m\,\Delta x + \Delta x \cdot o(\Delta x) \quad (\Delta x \to 0), \tag{4}$$

in which case $G'(x_0) = m$; cf. [2, Theorem 2].

Lin-Integ.

$$\forall u,\ v,\ w \in [a,\ b]:\ A_u^w = A_u^v + A_v^w. \tag{5}$$

Int-Mono.

$$f \le g \text{ on } [a,\ b] \Rightarrow \int_a^b f \le \int_a^b g. \tag{6}$$

Similarity. For $b > 0$,

$$\int_b^{ab} h(t)\,\mathrm{d}t = \int_1^a h(bu)\,b\,\mathrm{d}u. \tag{7}$$

Inv-Cont.

$$t \mapsto 1/t \in C^0(\mathbb{R}_{>0}). \tag{8}$$

Image-Union.

$$f(A\cup B) = f(A)\cup f(B). \tag{9}$$

Product rule.

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x). \tag{10}$$

Chain rule.

$$(f\circ g)'(x) = f'(g(x))\,g'(x). \tag{11}$$

Logarithm. For $x > 0$, set

$$\ln(x) := \int_1^x \frac{\mathrm{d}t}{t}. \tag{12}$$

By Equation 8 the integrand $t \mapsto 1/t$ is continuous on $\mathbb{R}_{>0}$, so the integral is well defined.

Sign-Law.

$$\forall u > 0:\ (\ln u < 0 \iff u < 1) \wedge (\ln u = 0 \iff u = 1) \wedge (\ln u > 0 \iff u > 1). \tag{13}$$

Ln-Bij. The map $\ln : \mathbb{R}_{>0} \to \mathbb{R}$ of Equation 12 is bijective; the *exponential* is defined as its inverse:

$$\ln : \mathbb{R}_{>0} \to \mathbb{R} \text{ bijective}, \qquad \exp := \ln^{-1} : \mathbb{R} \to \mathbb{R}_{>0}. \tag{14}$$

Equivalently, $\ln(\exp u) = u$ for all $u \in \mathbb{R}$ and $\exp(\ln a) = a$ for all $a > 0$.

Barrow-tangency. Let G be differentiable at x_0, and let $L(u) = m(u-x_0)+G(x_0)$ be the line through $(x_0,\ G(x_0))$ of slope m. If $L \le G$ (or $L \ge G$) on a punctured neighbourhood of

x_0, then[1]

$$m = G'(x_0). \tag{15}$$

2. Statements of the classical theorems

2.1. Topological theorems

Theorem 1 (Extreme Value Theorem).

If $f : [a, b] \to \mathbb{R}$ is continuous, then

$$\exists x_{\max}, x_{\min} \in [a, b], \ \forall x \in [a, b] : \ f(x_{\min}) \le f(x) \le f(x_{\max}). \tag{16}$$

Theorem 2 (Intermediate Value Theorem).

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then

$$\exists c \in (a, b) : \ f(c) = 0. \tag{17}$$

2.2. Differentiation theorems

Chord and tangent lines. For $f : [a, b] \to \mathbb{R}$ and distinct $u, v \in [a, b]$, the *chord line of f through u and v* is the unique affine line through the points $(u, f(u))$ and $(v, f(v))$; in coordinates,

$$d_{u,v} := \left\{(x, y) \in \mathbb{R}^2 : \ y = \frac{f(v) - f(u)}{v - u}\,(x - u) + f(u)\right\}. \tag{18}$$

For $x_0 \in (a, b)$ at which f is differentiable, the *tangent line to f at x_0* is

$$d_{x_0} := \{(x, y) \in \mathbb{R}^2 : \ y = f'(x_0)\,(x - x_0) + f(x_0)\}. \tag{19}$$

Two such lines are *parallel*, written $(d_1) \parallel (d_2)$, iff they have the same slope; in particular, $(d_c) \parallel (d_{a,b})$ iff $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.

Theorem 3 (Fermat).

If f has a local extremum at $c \in (a, b)$ and is differentiable at c, then

$$f'(c) = 0. \tag{20}$$

Theorem 4 (Rolle).

If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then

$$\exists c \in (a, b) : \ f'(c) = 0. \tag{21}$$

Theorem 5 (Mean-Value Theorem).

If f is continuous on $[a, b]$ and differentiable on (a, b), then

$$\exists c \in (a, b) : \ f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{22}$$

Geometrically, $(d_c) \parallel (d_{a,b})$.

[1] The two-sided hypothesis strengthens Barrow's one-sided criterion, which fails at inflections [2]. Every use in Part 1 is in a convex or concave setting, where the two-sided hypothesis holds.

Theorem 6 (Constant-function).
If f is continuous on $[a,\,b]$, differentiable on $(a,\,b)$, and $f'=0$ on $(a,\,b)$, then

$$f(x)=f(a)\quad \text{for all } x\in[a,\,b]. \tag{23}$$

Theorem 7 (Monotonicity).
If f is continuous on $[a,\,b]$, differentiable on $(a,\,b)$, and $f'\geq 0$ (resp. $f'>0$), then

$$x<y\Rightarrow f(x)\leq f(y)\quad (\text{resp. } f(x)<f(y)) \tag{24}$$

for all $x,\,y\in[a,\,b]$.

Theorem 8 (Cauchy MVT).
If $f,\,g$ are continuous on $[a,\,b]$, differentiable on $(a,\,b)$, and $g'\neq 0$ on $(a,\,b)$, then

$$\exists c\in(a,\,b):\ \frac{f'(c)}{g'(c)}=\frac{f(b)-f(a)}{g(b)-g(a)}. \tag{25}$$

Theorem 9 (Darboux).
If f is differentiable on $[a,\,b]$ and $f'(a)<\lambda<f'(b)$, then

$$\exists c\in(a,\,b):\ f'(c)=\lambda. \tag{26}$$

Theorem 10 (L'Hôpital).
Let $f,\,g$ be differentiable near a, with $f(a)=g(a)=0, g'\neq 0$ near a, and

$$\lim_{x\to a} f'(x)/g'(x)=L.$$

Then

$$\lim_{x\to a} f(x)/g(x)=L. \tag{27}$$

Theorem 11 (Taylor, Lagrange remainder).
If f is $(n+1)$-times differentiable on an interval containing a and x, then

$$\exists c \text{ between } a,\,x:\ f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k+\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}. \tag{28}$$

Theorem 12 (Convexity).
If f is twice differentiable on $[a,\,b]$ and $f''\geq 0$ on $(a,\,b)$, then

$$f(x)\geq f(x_0)+f'(x_0)(x-x_0)\quad \text{for all } x,\,x_0\in[a,\,b]. \tag{29}$$

Theorem 13 (Inverse function).
If $f: I\to\mathbb{R}$ is continuous and strictly monotone on I, x_0 lies in the interior I° of I, and f is differentiable at x_0 with $f'(x_0)\neq 0$, then f^{-1} is differentiable at $y_0=f(x_0)$ with

$$(f^{-1})'(y_0)=1/f'(x_0). \tag{30}$$

2.3. Integration theorems

Lemma 14 (Barrow's area inequality).

If f is continuous and increasing on $[a,\,b]$ and $x_0,\ x_0+\mathrm{d}x\in[a,\,b],\mathrm{d}x\neq 0$, then

$$f(x_0)\,\mathrm{d}x\le A_{x_0}^{x_0+\mathrm{d}x}\le f(x_0+\mathrm{d}x)\,\mathrm{d}x, \tag{31}$$

and both inequalities are strict when f is strictly increasing on the interval with endpoints x_0 and $x_0+\mathrm{d}x$.

Lemma 15 (MVT for integrals).

If f is continuous on $[a,\,b]$ and $x_0,\ x_0+\mathrm{d}x\in[a,\,b],\mathrm{d}x\neq 0$, then

$$\exists\theta\text{ between }x_0,\ x_0+\mathrm{d}x:\ \ A_{x_0}^{x_0+\mathrm{d}x}=f(\theta)\,\mathrm{d}x. \tag{32}$$

Theorem 16 (First FTC; Barrow).

If f is continuous on $[a,\,b]$, then

$$\frac{d}{dx}\int_a^x f=f(x) \tag{33}$$

for every $x\in(a,\,b)$, with one-sided versions at $a,\,b$.

Theorem 17 (Second FTC).

If F is differentiable on $[a,\,b]$ and $F'=f$ is continuous, then

$$\int_a^b f=F(b)-F(a). \tag{34}$$

Theorem 18 (Integration by parts).

If $f,\,g\in C^1([a,\,b])$, then

$$\int_a^b fg'=[fg]_a^b-\int_a^b f'g. \tag{35}$$

Theorem 19 (Change of variables).

If $g\in C^1([a,\,b])$ and f is continuous on $g([a,\,b])$, then

$$\int_{g(a)}^{g(b)} f=\int_a^b f(g)\,g'. \tag{36}$$

Theorem 20 (Weighted MVT).

If f is continuous on $[a,\,b],w\ge 0$ is Riemann integrable, and $\int_a^b w>0$, then

$$\exists c\in[a,\,b]:\ \int_a^b fw=f(c)\int_a^b w. \tag{37}$$

2.4. The logarithm and the exponential

Theorem 21 (Log-Property).

With $\ln(x):=\int_1^x \mathrm{d}t/t$ $(x>0)$,

$$\forall a,\,b>0:\ \ln(a)+\ln(b)=\ln(ab). \tag{38}$$

Theorem 22 (Exp-Property).
With $\exp := \ln^{-1}$,

$$\forall u,\, v \in \mathbb{R} :\ \exp(u)\exp(v) = \exp(u+v). \tag{39}$$

Theorem 23 (Exp-Image).

$$\exp(\mathbb{R}) = \mathbb{R}_{>0}, \qquad \ln(\mathbb{R}_{>0}) = \mathbb{R}. \tag{40}$$

3. Geometrical proofs

3.1. Topological theorems

Proof of Theorem 1.

Set $S = f([a,b])$; S is nonempty and bounded. Let

$$M = \sup S, \qquad m = \inf S.$$

Were M unattained, the function $1/(M-f)$ would be continuous and unbounded on $[a,b]$, a contradiction. Hence

$$M = f(x_{\max})$$

for some $x_{\max} \in [a,b]$; symmetrically $m = f(x_{\min})$. Combining,

$$f(x_{\min}) \le f(x) \le f(x_{\max}) \qquad (x \in [a,b]),$$

which is Equation 16. □

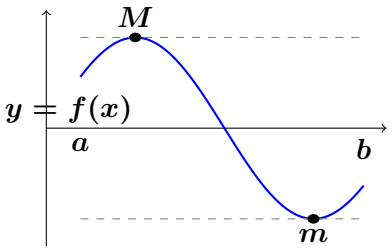


Figure 1. EVT.

Proof of Theorem 2.

Assume $f(a) < 0 < f(b)$ and set

$$c = \sup\{x \in [a,b] : f(x) < 0\}.$$

By continuity, $f(c) \le 0$. If $f(c) < 0$, then $f < 0$ on a neighbourhood of c intersected with $[a,b]$, contradicting the definition of c as an upper bound. Hence

$$f(c) = 0,$$

which is Equation 17. □

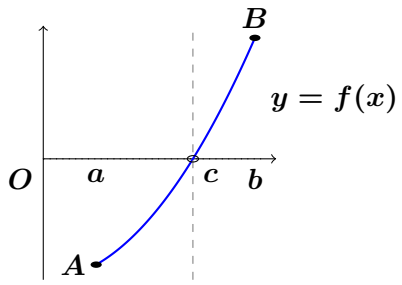


Figure 2. IVT.

Proof of Theorem 3.

Assume c is a local maximum; the local-minimum case is symmetric. For $h > 0^+$ small,

$$\frac{f(c+h)-f(c)}{h} \le 0 \quad \Longrightarrow \quad f'(c) \le 0;$$

for $h < 0^-$ small,

$$\frac{f(c+h)-f(c)}{h} \ge 0 \quad \Longrightarrow \quad f'(c) \ge 0.$$

Hence

$$f'(c) = 0.$$

□

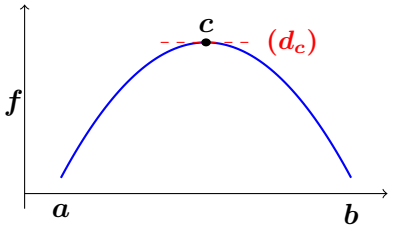


Figure 3. Fermat.

Proof of Theorem 4.

By EVT, f attains $M = \max_{[a,b]} f$ and $m = \min_{[a,b]} f$. If $M = m$, then f is constant and

$$f'(c) = 0 \qquad (c \in (a,b)).$$

Otherwise $f(a) = f(b)$ forces an interior extremum $c \in (a,b)$, and Fermat's theorem gives

$$f'(c) = 0.$$

□

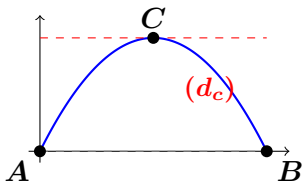


Figure 4. Rolle.

Proof of Theorem 5 (Barrow's sliding tangent).

Let L be the affine function whose graph is the chord line $(d_{a,b})$ of Equation 18, so that $L' \equiv (f(b) - f(a))/(b - a)$; set $g := f - L$. Then $g(a) = g(b) = 0$. By EVT, g attains $M = \max g$ and $m = \min g$. *Case $M > 0$:* the maximum is attained at some $c \in (a, b)$; the translate $L + M$ touches f at C and lies above, so by concave Barrow-tangency,

$$f'(c) = L'(c) = \frac{f(b) - f(a)}{b - a}.$$

Case $M = 0$: if $f \equiv L$ the result is trivial; otherwise $m < 0$ with minimum at interior c, and the same argument via convex Barrow-tangency yields

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

In either case, $(d_c) \parallel (d_{a,b})$. □

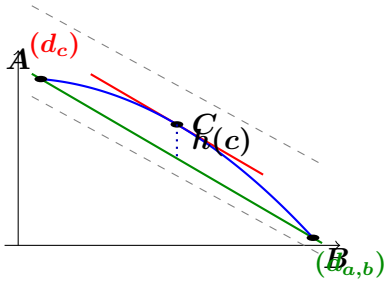


Figure 5. MVT sliding tangent: chord $(d_{a,b})$ (green) and tangent (d_c) (red), with $(d_c) \parallel (d_{a,b})$.

Proof of Theorem 6.

Apply MVT on $[a, x]$ with $x \in (a, b]$:

$$f(x) - f(a) = f'(c)(x - a) = 0.$$

Hence

$$f(x) = f(a) \qquad (x \in [a, b]).$$

□

Proof of Theorem 7.

Let $x_1 < x_2$ in $[a, b]$. By MVT on $[x_1, x_2]$, there is $c \in (x_1, x_2)$ with

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

If $f' \geq 0$, the right-hand side is ≥ 0, hence $f(x_2) \geq f(x_1)$; if $f' > 0$, it is > 0, hence $f(x_2) > f(x_1)$. □

Proof of Theorem 8.

Define

$$h := (f(b) - f(a))\, g - (g(b) - g(a))\, f.$$

Then $h(a) = h(b)$, so Rolle's theorem gives $c \in (a, b)$ with

$$(f(b) - f(a))\, g'(c) = (g(b) - g(a))\, f'(c).$$

Since g' is nonzero on (a, b) and $g(b) \neq g(a)$, dividing yields

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

□

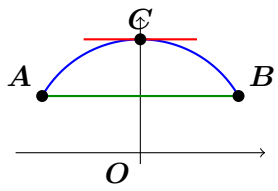


Figure 6. Cauchy MVT.

Proof of Theorem 9.

Set $g := f - \lambda x$; then

$$g'(a) = f'(a) - \lambda < 0, \qquad g'(b) = f'(b) - \lambda > 0.$$

For small $h > 0$,

$$g(a + h) < g(a), \qquad g(b - h) < g(b).$$

By EVT, g attains its minimum at some $c \in [a, b]$; the above inequalities exclude $c = a$ and $c = b$, so $c \in (a, b)$. By Fermat's theorem,

$$g'(c) = 0, \qquad \text{i.e.} \qquad f'(c) = \lambda.$$

□

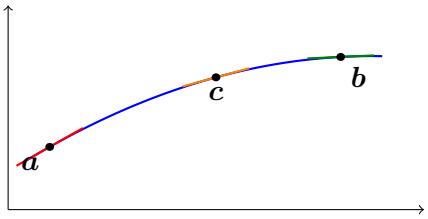


Figure 7. Darboux.

Proof of Theorem 10.

By Cauchy MVT on $[a, x]$ with $x > a$,

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}, \qquad c_x \in (a, x).$$

Since $c_x \to a$ as $x \to a^+$,

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = \lim_{x\to a^+} \frac{f'(c_x)}{g'(c_x)} = L.$$

The case $x \to a^-$ is symmetric, so the two-sided limit is L. □

Proof of Theorem 11.

Let

$$P(t) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(t-a)^k, \qquad M := \frac{f(x) - P(x)}{(x-a)^{n+1}},$$

and define $h(t) := f(t) - P(t) - M(t-a)^{n+1}$. Then

$$h^{(j)}(a) = 0 \quad (j = 0, \dots, n), \qquad h(x) = 0.$$

Applying Rolle's theorem $n+1$ times gives c strictly between a and x with

$$h^{(n+1)}(c) = 0.$$

Since $h^{(n+1)}(t) = f^{(n+1)}(t) - M\,(n+1)!$, this yields

$$M = \frac{f^{(n+1)}(c)}{(n+1)!},$$

hence

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}\,(x-a)^{n+1}.$$

□

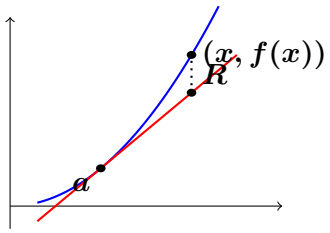


Figure 8. Taylor $(n = 1)$.

Proof of Theorem 12.

Set $g(x) := f(x) - f(x_0) - f'(x_0)(x - x_0)$. Then

$$g'' = f'' \geq 0, \qquad g'(x_0) = 0, \qquad g(x_0) = 0.$$

Since g' is nondecreasing and $g'(x_0) = 0$, we have $g'(x) \leq 0$ for $x < x_0$ and $g'(x) \geq 0$ for $x > x_0$. By the monotonicity criterion, g is nonincreasing on $[a, x_0]$ and nondecreasing on $[x_0, b]$. With $g(x_0) = 0$,

$$g(x) \geq 0 \qquad (x \in [a, b]),$$

i.e.

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0).$$

□

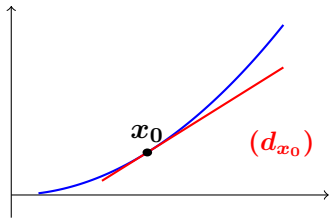


Figure 9. Convexity.

Proof of Theorem 13.

Let $c := f'(x_0) \neq 0$. Since $x_0 \in I^\circ$, choose $\rho > 0$ with $(x_0 - \rho, x_0 + \rho) \subseteq I$.

Local surjectivity. Assume w.l.o.g. that f is strictly increasing on I (the strictly decreasing case is symmetric). Choose $x_0^-, x_0^+ \in (x_0 - \rho, x_0 + \rho)$ with $x_0^- < x_0 < x_0^+$. Then

$$f(x_0^-) < y_0 < f(x_0^+).$$

By IVT on the closed subinterval $[x_0^-, x_0^+] \subseteq I$, every $y \in (f(x_0^-), f(x_0^+))$ is attained at some $x \in (x_0^-, x_0^+) \subseteq I$. Hence $(f(x_0^-), f(x_0^+)) \subseteq f(I)$ is an open neighbourhood of y_0, and f^{-1} is defined on it.

Continuity of f^{-1} at y_0:

Differentiability of f at x_0 gives $\delta_0 \in (0, \rho)$ such that for all x with $|x - x_0| < \delta_0$,

$$|f(x) - f(x_0) - c(x - x_0)| < \tfrac{|c|}{2} |x - x_0|,$$

whence by the reverse triangle inequality

$$|f(x) - f(x_0)| \geq \tfrac{|c|}{2} |x - x_0|.$$

Given $\varepsilon > 0$, set $\varepsilon' := \min(\varepsilon, \delta_0)$ and

$$\eta := \min\Big(\tfrac{|c|}{2}\,\varepsilon',\ y_0 - f(x_0^-),\ f(x_0^+) - y_0\Big) > 0.$$

Suppose $|y - y_0| < \eta$.
Since $y \in (f(x_0^-), f(x_0^+)) \subseteq f(I)$, the value $x := f^{-1}(y) \in (x_0 - \rho, x_0 + \rho) \subseteq I$ is defined.
In the case $|x - x_0| < \delta_0$, the bound above gives $|y - y_0| \geq \frac{|c|}{2}|x - x_0|$, so $|x - x_0| < \varepsilon' \leq \varepsilon$.
In the case $|x - x_0| \geq \delta_0$, let $x_1 := x_0 + \operatorname{sgn}(x - x_0)\,\delta_0 \in (x_0 - \rho, x_0 + \rho) \subseteq I$.
By strict monotonicity of f, $|y - y_0| \geq |f(x_1) - f(x_0)|$;
passing the bound $|f(u) - f(x_0)| \geq \frac{|c|}{2}|u - x_0|$ to the limit $u \to x_1$ from inside (x_0, x_1) yields

$$|f(x_1) - f(x_0)| \geq \tfrac{|c|}{2}\,\delta_0 \geq \tfrac{|c|}{2}\,\varepsilon' \geq \eta,$$

contradicting $|y - y_0| < \eta$. Hence $|y - y_0| < \eta \Rightarrow |f^{-1}(y) - x_0| < \varepsilon$, so f^{-1} is continuous at y_0.

Derivative of f^{-1} at y_0:

Changing variable $y = f(x)$, i.e. $x = f^{-1}(y) \to x_0$ as $y \to y_0$ (continuity of f^{-1} at y_0 just established justifies the substitution),

$$(f^{-1})'(y_0) = \lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \lim_{x \to x_0} \frac{x - x_0}{f(x) - f(x_0)} = \frac{1}{f'(x_0)},$$

which is Equation 30. □

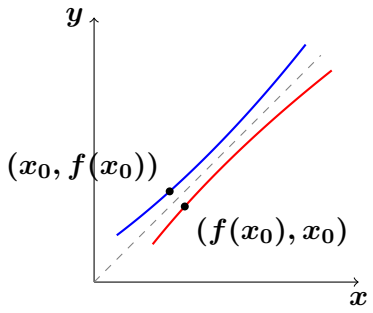


Figure 10. Inverse.

3.2. Integration theorems

Proof of Theorem 14.

For $\mathrm{d}x > 0$, $f(x_0) \leq f \leq f(x_0 + \mathrm{d}x)$ on the interval, so inscribed/circumscribed rectangles bound A:

$$f(x_0)\,\mathrm{d}x \leq A_{x_0}^{x_0 + \mathrm{d}x} \leq f(x_0 + \mathrm{d}x)\,\mathrm{d}x.$$

For $\mathrm{d}x < 0$, apply Equation 2 to reverse both sides. □

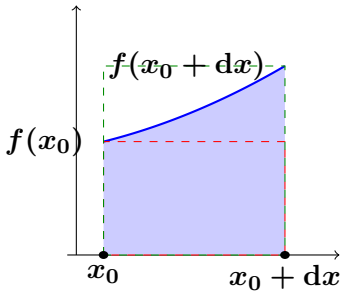


Figure 11. Barrow's area inequality.

Proof of Theorem 15.

Define $G(t) := A_{x_0}^{t}$. By FFTC, G is differentiable with $G' = f$.

MVT on the interval with endpoints x_0 and $x_0 + \mathrm{d}x$ gives θ between them with

$$G'(\theta) = \frac{G(x_0 + \mathrm{d}x) - G(x_0)}{\mathrm{d}x}.$$

Since $G(x_0) = 0$ and $G(x_0 + \mathrm{d}x) = A_{x_0}^{x_0+\mathrm{d}x}$,

$$A_{x_0}^{x_0+\mathrm{d}x} = f(\theta)\,\mathrm{d}x.$$

□

Proof of Theorem 16.

Fix $x_0 \in (a, b)$ and $\varepsilon > 0$. For $h \neq 0$ with $x_0 + h \in [a, b]$, let I_h be the closed interval with endpoints x_0 and $x_0 + h$. Since f is continuous at x_0, for $|h|$ small there exist $m_\varepsilon, M_\varepsilon$ with

$$m_\varepsilon \leq f(u) \leq M_\varepsilon \ (u \in I_h), \qquad |m_\varepsilon - f(x_0)| < \varepsilon, \qquad |M_\varepsilon - f(x_0)| < \varepsilon.$$

By integral monotonicity applied on I_h to the constants $m_\varepsilon, M_\varepsilon$, for $h > 0$,

$$m_\varepsilon h \leq A_{x_0}^{x_0+h} \leq M_\varepsilon h;$$

for $h < 0$, the same inequality holds with both directions reversed (the oriented convention Equation 2 changes the sign of both $A_{x_0}^{x_0+h}$ and h). In either case, dividing by h preserves the direction, giving

$$m_\varepsilon \leq \frac{A_a^{x_0+h} - A_a^{x_0}}{h} \leq M_\varepsilon,$$

where the numerator uses additivity Equation 5.

Since $\varepsilon > 0$ was arbitrary and $m_\varepsilon, M_\varepsilon \to f(x_0)$ as $\varepsilon \to 0$, the squeeze theorem yields

$$\frac{d}{dx} A_a^x \Big|_{x_0} = f(x_0).$$

As $x_0 \in (a, b)$ was arbitrary, Equation 33 holds on (a, b); the one-sided cases follow by one-sided limits. □

Remark (Barrow's original argument). Barrow's proof of the FFTC [1, Lecture X, §11] (see [2, §2–3]) assumes *additionally* that f is *increasing*, which makes the rectangle estimate immediate:

for $\mathrm{d}x > 0$,

$$f(x_0)\,\mathrm{d}x \le A_{x_0}^{x_0+\mathrm{d}x} \le f(x_0+\mathrm{d}x)\,\mathrm{d}x,$$

and both inequalities reverse for $\mathrm{d}x < 0$ (by Equation 2).
Barrow then shows that the line

$$L(u) = f(x_0)(u - x_0) + A_a^{x_0},$$

satisfies

$$L(x_0 + \mathrm{d}x) \le A_a^{x_0+\mathrm{d}x}$$

(respectively $\ge$ for $\mathrm{d}x < 0$) and invokes the tangency principle Equation 15 to conclude

$$f(x_0) = dA_a^x/dx|_{x_0}.$$

The proof above removes the monotone assumption and uses only continuity via the squeeze $m_\varepsilon \le (A_a^{x_0+h} - A_a^{x_0})/h \le M_\varepsilon$; the conclusion is the same.

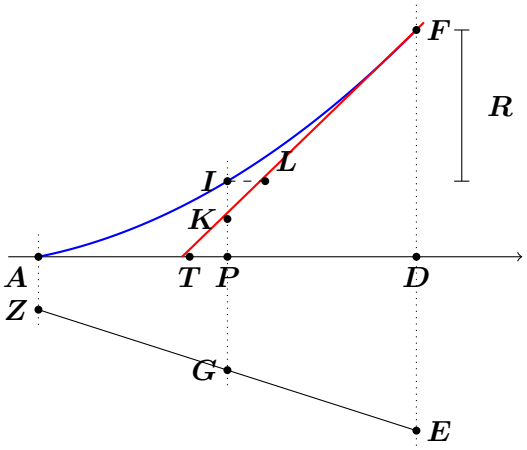


Figure 12. Barrow's Fig. 109.

Proof of Theorem 17.

Define $G(x) := \int_a^x f$. By FFTC, $G' = f$, so $(F - G)' = 0$ and hence $F - G$ is constant. Therefore

$$F(b) - F(a) = \int_a^b f.$$

□

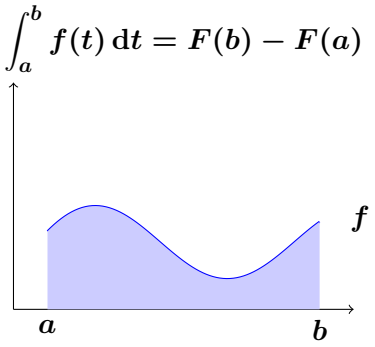


Figure 13. Second FTC.

Proof of Theorem 18.

The product rule gives

$$(fg)' = f'g + fg'.$$

Integrating with the Second FTC,

$$[fg]_a^b = \int_a^b (fg)' = \int_a^b f'g + \int_a^b fg'.$$

Rearranging,

$$\int_a^b f\,g' = [fg]_a^b - \int_a^b f'\,g.$$

□

Proof of Theorem 19.

Let $F' = f$. By the chain rule,

$$(F \circ g)' = f(g)\,g'.$$

Applying the Second FTC on both sides gives

$$\int_a^b f(g)\,g' = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f.$$

□

Proof of Theorem 20.

Since $m \le f \le M$ and $w \ge 0$,

$$m\int_a^b w \le \int_a^b fw \le M\int_a^b w.$$

So $A := \int_a^b fw / \int_a^b w \in [m, M]$; by IVT,

$$f(c) = A,$$

i.e. $\int_a^b fw = f(c)\int_a^b w$. □

3.3. The logarithm and the exponential

Proof of Theorem 21.

We have

$$\ln(ab) - \ln(b) = \int_b^{ab} \frac{\mathrm{d}t}{t}.$$

The substitution $(t, y) \mapsto (t/b, by)$ preserves both area and the identity $y = 1/t$, so

$$\int_b^{ab} \frac{\mathrm{d}t}{t} = \int_1^a \frac{\mathrm{d}u}{u} = \ln(a).$$

Hence $\ln(ab) - \ln(b) = \ln(a)$, i.e. $\ln(a) + \ln(b) = \ln(ab)$. □

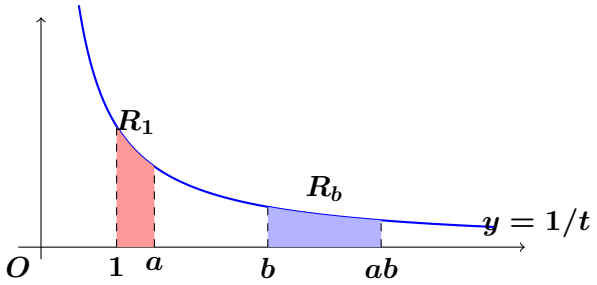


Figure 14. Hyperbolic similarity.

Proof of Theorem 22.

Since ln is injective, it suffices to show

$$\ln(\exp(u)\exp(v)) = \ln(\exp u) + \ln(\exp v) = u + v = \ln(\exp(u+v)).$$

Injectivity of ln then gives

$$\exp(u)\ \exp(v) = \exp(u+v).$$

□

Proof of Theorem 23.

The Sign-Law gives

$$\exp(\mathbb{R}_{<0}) =]0,1[, \qquad \exp(\{0\}) = \{1\}, \qquad \exp(\mathbb{R}_{>0}) =]1,+\infty[.$$

The Image-Union law then gives $\exp(\mathbb{R}) = \mathbb{R}_{>0}$, and bijectivity of ln gives $\ln(\mathbb{R}_{>0}) = \mathbb{R}$. □

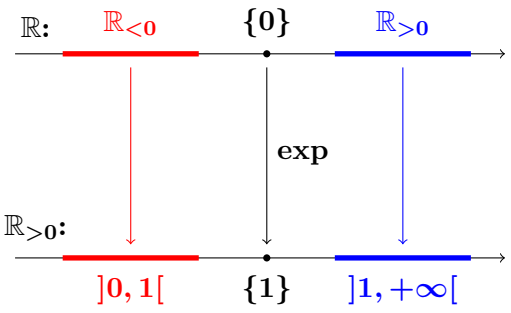


Figure 15. Image of Exponential Function exp

4. Remarks on the geometric development

Barrow's sliding-tangent MVT proof is retained for its historical content; the Barrow-tangency principle Equation 15 is eliminable via Rolle and Monotone + MVT. The FFTC proof uses only area additivity Equation 5 and rectangle comparison Equation 31; no Riemann-sum limit is invoked, as emphasized by Barrow [1] and confirmed in Part 2.

Dependency summary

EVT, IVT → Fermat → Rolle → MVT → Const, Monotone, Cauchy, Darboux, Taylor, Convexity, Inverse. Second FTC from FFTC + Const. IBP, Subst, Weighted MVT from Second FTC or IVT/IntMono. Log from IntAdd + Similarity; Exp from Log; Image from LnBij + Sign-Law + ImgUnion. Barrow-area + MVTInt → FFTC → Log.

Part 2

Reverse-Mathematical Analysis in RCA_0

Every Part 1 entry reappears as a tagged statement with a dependency table and an exact strength in {FOL, RCA_0, WKL_0, ACA_0, ZF, ZFC}. Geometric proofs are at most WKL_0 as written; §11 gives the matching RCA_0-strength tables via Hardin–Velleman, Yokoyama, or direct argument.

5. Coding conventions

Following Yokoyama [6], we distinguish:

- ***Pointwise differentiability:*** **the limit exists for each $x \in I$; no derivative function needed. Hardin–Velleman's Rolle and MVT in RCA_0 use this.**
- ***Strong C^k-coding:*** $C^k_{\text{strong}}(I)(f) :\iff \exists g_0, \ldots, g_k : \; g_0 = f \wedge \forall j < k, \; g_{j+1} = g_j'$. **All C^k hypotheses in Part 2 mean strong C^k-coding.**

- ***Inverse function:*** **Yokoyama [6, Thm. 3.22] in RCA_0 for strong C^1-functions with $f' \neq 0$ on $I, x_0 \in I^\circ$. The pointwise form is not RCA_0-provable.**

6. Research context

6.1. The core hierarchy

RCA_0 = recursive comprehension + Σ^0_1-induction; no choice.

Result	Strength	Reference	Choice
MVT	RCA_0	[3]	none
IVT, EVT	WKL_0	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[4, 5]	none
ln/exp chain	RCA_0	[6]	none

Key facts:

- **MVT $\in \mathsf{RCA}_0$ (Hardin–Velleman [3]); no EVT, no IVT.**
- **IVT $\equiv \mathsf{WKL}_0$: sole non-RCA_0 ingredient is $\sup\{x : f(x) < 0\}$.**
- **Inverse function $\in \mathsf{RCA}_0$ (Yokoyama [6, Thm. 3.22]).**
- **ln/exp $\in \mathsf{RCA}_0$: definitional plus integral postulates.**

Remark 1 (Proofs vs. classification).

Part 1** proofs use* **sup, inf, *LUB (ax:R-complete). MVT is RCA_0 via Hardin–Velleman. IVT, EVT are WKL_0-equivalent over RCA_0 and remain WKL_0.

6.2. Illustrative theorems

Theorem	Strength	Reference	Choice
IVT	WKL_0	[4]	none
MVT	RCA_0	[3]	none
Hahn–Banach	$\equiv \mathsf{WKL}_0$	[4]	none
Weierstrass approx.	$\equiv \mathsf{WKL}_0$	[4]	none
Peano existence	$\equiv \mathsf{WKL}_0$	[4]	none
sequential IVT/MVT	$\equiv \mathsf{WKL}_0$	[5]	none
separation	$\equiv \mathsf{ACA}_0$	[15]	none
Banach–Alaoglu	ZFC	[12]	full AC
Tychonoff	ZFC	[13]	full AC
well-ordering	ZFC	[14]	full AC

Strictness: $\mathsf{RCA}_0 \subsetneq \mathsf{WKL}_0 \subsetneq \mathsf{ACA}_0 \subsetneq$ ZFC.

6.3. Dedekind vs. Cauchy completeness

- **Dedekind (LUB): every nonempty bounded-above subset has a supremum. A ZF theorem; equivalent over ordered fields to IVT, EVT, MVT, Darboux [7, 8]. Form used here: ax:R-complete.**

- Cauchy completeness: every Cauchy sequence converges. ZF-equivalent to $\mathsf{DC}_{\mathbb{R}}$. Not a ZF theorem. Not used here.

Solovay's model [16]: under ZF+DC, all reals are Lebesgue measurable and AC fails; $\mathsf{DC} \Rightarrow \mathsf{DC}_{\mathbb{R}}$, so Cauchy completeness holds there. The choice-free classification is strictly finer than $\mathsf{DC}_{\mathbb{R}}$.

6.4. Constructive RM

In BISH, IVT $\equiv$ LLPO [9]. This document is classical.

6.5. Lean 4 and Mathlib

Lean 4 and Mathlib [10]: about 70.4% of declarations depend syntactically on Classical.choice, of which roughly half reach it only through classical logic (law of excluded middle, propext, Quot.sound) rather than through genuine choice; 13.1% have choice-free statements with choice-dependent proofs [11]. Our classification is stronger: it certifies choice-free proofs, not just statements.

7. Axioms

The fourteen axioms of $\mathbb{R}$ as a complete ordered field.

Ax 1	ax:add-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x + y) + z = x + (y + z)$	Add.Assoc.	ZF
Ax 2	ax:add-comm : $\forall(x,\, y : \mathbb{R}),\ \ x + y = y + x$	Add.Comm.	ZF
Ax 3	ax:add-zero : $\forall(x : \mathbb{R}),\ \ 0 + x = x$	Add.Id.	ZF
Ax 4	ax:add-inv : $\forall(x : \mathbb{R}),\ \ x + (-x) = 0$	Add.Inv.	ZF
Ax 5	ax:mul-assoc : $\forall(x,\, y,\, z : \mathbb{R}),\ \ (x \cdot y) \cdot z = x \cdot (y \cdot z)$	Mul.Assoc.	ZF
Ax 6	ax:mul-comm : $\forall(x,\, y : \mathbb{R}),\ \ x \cdot y = y \cdot x$	Mul.Comm.	ZF
Ax 7	ax:mul-one : $\forall(x : \mathbb{R}),\ \ 1 \cdot x = x$	Mul.Id.	ZF
Ax 8	ax:mul-inv : $\forall(x : \mathbb{R}),$ $(x \neq 0 \to x \cdot x^{-1} = 1) \wedge (x = 0 \to x^{-1} = 0)$	Mul.Inv.	ZF
Ax 9	ax:distrib : $\forall(x,\, y,\, z : \mathbb{R}),\ \ x \cdot (y + z) = x \cdot y + x \cdot z$	Distrib.	ZF
Ax 10	ax:nontriv : $0 \neq 1$	Nontriv.	ZF
Ax 11	ax:order : $\big(\forall x :\ x \le x\big) \wedge \big(\forall x,$ $y :\ x \le y \wedge y \le x \to x = y\big) \wedge \big(\forall x,\, y,$ $z :\ x \le y \wedge y \le z \to x \le z\big) \wedge \big(\forall x,\, y :\ x \le y \vee y \le x\big)$	LinearOrder	ZF
Ax 12	ax:add-mon : $\forall(x,\, y,\, z : \mathbb{R}),\ \ x \le y \to x + z \le y + z$	AddMon.	ZF
Ax 13	ax:mul-mon : $\forall(x,\, y : \mathbb{R}),\ \ 0 \le x \wedge 0 \le y \to 0 \le x \cdot y$	MulMon.	ZF
Ax 14	ax:complete : $\big(\exists x,\, S(x)\big) \wedge \big(\exists u,\, \forall x,$ $S(x) \to x \le u\big) \to \exists s,\, \big(\forall x,\, S(x) \to x \le s\big) \wedge \big(\forall u,\, (\forall x,$ $S(x) \to x \le u) \to s \le u\big)$	Cut	ZF

The fourteen axioms of $\mathbb{R}$ as a complete ordered field are listed above. Only the first thirteen are used in Part 2; the fourteenth, ax:R-complete (Cut), enters no Part 2 proof. It is ZF-realizable via Dedekind cuts without choice.

8. Postulates

Unproved internal statements. Each is RCA_0-provable or first-order logical; none introduces choice.

8.1. Integral postulates

Pos 1	**Pos:IntLinear** : $\forall(a,b:\mathbb{R}),\ \forall(f,g\in L^1([a,b])),\ \forall(\alpha,\beta:\mathbb{R}),\ \int_a^b(\alpha f(x)+\beta g(x))\,dx=\alpha\int_a^b f(x)\,dx+\beta\int_a^b g(x)\,dx$	**IntLin**	RCA_0
Pos 2	**Pos:IntAdd** : $\forall(a,b,c:\mathbb{R}),\ \forall(f\in L^1([\min(a,b,c),\max(a,b,c)])),\ \int_a^b f(x)\,dx=\int_a^c f(x)\,dx+\int_c^b f(x)\,dx$	**IntAdd**	RCA_0

Note: $f\in L^1([\alpha,\beta])$ and $[\gamma,\delta]\subseteq[\alpha,\beta]$ imply $f\in L^1([\gamma,\delta])$; cited as `restrict` in proof tables.

Pos 3	**Pos:IntMono** : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f,g\in L^1([a,b])),\ \big(\forall(x\in[a,b]),\ f(x)\le g(x)\big)\to\int_a^b f(x)\,dx\le\int_a^b g(x)\,dx$	**IntMono**	RCA_0
Pos 4	**Pos:IntSubst** : $\forall(a,b:\mathbb{R}),\ \forall(g:\mathbb{R}\to\mathbb{R}),\ \forall(f:\mathbb{R}\to\mathbb{R}),\ g\in C^1([\min(a,b),\max(a,b)])\wedge\big(\forall t\in[\min(a,b),\max(a,b)],\ g'(t)\neq0\big)\wedge f\in C^0(g([\min(a,b),\max(a,b)]))\to\int_{g(a)}^{g(b)}f(x)\,dx=\int_a^b f(g(t))\,g'(t)\,dt$	**IntSubst**	RCA_0
Pos 5	**Pos:ContInt** : $\forall(a,b:\mathbb{R}),\ a<b\to\forall(f:\mathbb{R}\to\mathbb{R}),\ f\in C^0([a,b])\to f\in L^1([a,b])$	**ContInt**	RCA_0
Pos 6	**Pos:Similarity** : $\forall(a,b:\mathbb{R}),\ 0<a\wedge0<b\to\forall(h:\mathbb{R}\to\mathbb{R}),\ \int_b^{a\cdot b}h(t)\,dt=\int_1^a h(bu)\cdot b\,du$	**Sim.**	RCA_0

8.2. Differential postulates

Pos 7	**Pos:ProdRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x)\wedge\mathsf{Diff}(g,x)\to\mathsf{Diff}(f\cdot g,x)\wedge(f\cdot g)'(x)=f'(x)g(x)+f(x)g'(x)$	**Prod**	RCA_0
Pos 8	**Pos:ChainRule** : $\forall(f,g:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(g,x)\wedge\mathsf{Diff}(f,g(x))\to\mathsf{Diff}(f\circ g,x)\wedge(f\circ g)'(x)=f'(g(x))\cdot g'(x)$	**Chain**	RCA_0
Pos 9	**Pos:InvCont** : $\big(t\mapsto1/t\big)\in C^0(\mathbb{R}_{>0})$	**InvCont**	RCA_0
Pos 10	**Pos:BarrowTangency** : $\forall(G:\mathbb{R}\to\mathbb{R}),\ \forall(x_0,m:\mathbb{R}),\ \mathsf{Diff}(G,x_0)\wedge\Big(\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\le G(u)\big)\vee\big(\forall(u\neq x_0)\text{ near }x_0,\ G(x_0)+m(u-x_0)\ge G(u)\big)\Big)\to m=G'(x_0)$	**Barrow**	RCA_0

8.3. Logarithmic postulates

Pos 11	**Pos:LnBij** : $\mathsf{Bijective}(\ln:\mathbb{R}_{>0}\to\mathbb{R})$	**LnBij**	RCA_0
Pos 12	**Pos:LnSign** : $\forall(a:\mathbb{R}_{>0}),\ \big(\ln(a)<0\leftrightarrow a<1\big)\wedge\big(\ln(a)=0\leftrightarrow a=1\big)\wedge\big(\ln(a)>0\leftrightarrow a>1\big)$	**LnSign**	RCA_0

8.4. Logical postulate

Pos 13	**Pos:ImgUnion** : $\forall(X, Y : \mathsf{Type}),\ \forall(f : X \to Y),\ \forall(A, B : X \to \mathsf{Prop}),\ f(A \cup B) = f(A) \cup f(B)$	**ImgUnion**	**FOL**

8.5. External results

Pos 14	**Pos:ContImgBdd** : $\forall(f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \to \big(\exists L, U : \mathbb{R},\ \forall x : [a,b],\ L \le f(x) \le U\big)$	**BddImg**	**RCA$_0$**
Pos 15	**Pos:ContImgAttain** : $\forall(f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \to \big(\exists x_{\max} : [a,b], \forall x : [a,b], f(x) \le f(x_{\max})\big) \wedge \big(\exists x_{\min} : [a,b], \forall x : [a,b], f(x_{\min}) \le f(x)\big)$	**Attain**	**WKL$_0$**

Note on Pos:ContImgBdd and the coding of $C^0([a,b])$. Under the coding of §5 (following Simpson [4]), a continuous function $f : [a,b] \to \mathbb{R}$ carries a coded modulus of continuity, from which boundedness of the image $f([a,b])$ follows in RCA$_0$. The **Pos:ContImgBdd** tag RCA$_0$ reflects this standard coding convention. Under an alternative coding — pointwise continuity without a coded modulus — boundedness of continuous functions on $[a,b]$ rises to WKL$_0$; the choice between codings is a foundation choice, not a mathematical error, and §5 fixes the Simpson convention for this document.

Pos 16	**Pos:Π^0_1Sup** : $\forall(\varphi : \mathbb{N} \times \mathbb{R} \to \{0,1\}),\ \forall(a, b : \mathbb{R}),\ a \le b \to \big(\exists x \in [a,b], \forall n\, \varphi(n,x) = 0\big) \to \exists s \in [a,b],\ \big(\forall x \in [a,b], (\forall n\, \varphi(n,x) = 0) \to x \le s\big) \wedge \big(\forall u \in [a,b], (\forall x \in [a,b], (\forall n\, \varphi(n,x) = 0) \to x \le u) \to s \le u\big)$	**Sup**	**WKL$_0$**
Pos 17	**Pos:Σ^0_1Choice** : $\big(\forall n, \exists m, R(n,m)\big) \to \exists g, \forall n, R(n, g(n))$	Σ^0_1	**RCA$_0$**
Pos 18	**Pos:SeqCont** : $\forall(f : [a,b] \to \mathbb{R}), f \in C^0([a,b]) \to \forall x : [a,b],\ \forall(x_n : \mathbb{N} \to [a,b]),\ (x_n \to x) \to \big(f(x_n) \to f(x)\big)$	**SeqCont**	**RCA$_0$**
Pos 19	**Pos:HVRolle** : $\forall(f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge \big(\forall x : (a,b),\ \mathsf{Diff}(f,x)\big) \wedge f(a) = f(b) \to \exists c : (a,b),\ f'(c) = 0$	**HVRolle**	**RCA$_0$**
Pos 20	**Pos:YokoyamaInv** : $\forall(f : I \to \mathbb{R}),\ \forall(x_0 : I^\circ),\ C^1_{\mathbf{strong}}(I)(f) \wedge (\forall x : I, f'(x) \neq 0) \wedge \mathsf{StrictMono}(f) \to \mathsf{Diff}(f^{-1}, f(x_0)) \wedge (f^{-1})'(f(x_0)) = 1/f'(x_0)$	**YokInv**	**RCA$_0$**

9. Definitions

Definitional abbreviations available in RCA$_0$: set notation by Δ^0_1-comprehension; L^1, C^k, C^k_{strong} by Σ^0_1 predicates; A^x_a, ln, exp by unique existence + graphs; Diff, ChordSlope, TangentSlope, LocalExt, Increasing, StrictInc, StrictDec, StrictMono by definitional extension.

Def 1	**Def:R$_{>0}$** : $\mathbb{R}_{>0} := \{x : \mathbb{R} \mid x > 0\},\ \mathbb{R}_{<0} := \{x : \mathbb{R} \mid x < 0\},\ \mathbb{R}_{\ge 0} := \{x : \mathbb{R} \mid x \ge 0\}$	**R$_{>0}$**	**RCA$_0$**
Def 2	**Def:L^1** : $L^1([a,b]) := \{f : \mathbb{R} \to \mathbb{R} \mid \int_a^b |f(x)|\, dx < \infty\}$	**L^1**	**RCA$_0$**

Def 3	**Def:C^k** : $\forall(k:\mathbb{N}),\ \forall(U:\mathbb{R}\to\mathsf{Prop}),\ C^k(U) := \{f:\mathbb{R}\to\mathbb{R} \mid f \text{ k−timescont. diff. on } U\}$	**C^k**	**RCA$_0$**
Def 4	**Def:C$^k_{\text{strong}}$** : $\forall(k:\mathbb{N}),\ \forall(I:\mathbb{R}\to\mathsf{Prop}),\ C^k_{\text{strong}}(I)(f) :\iff \exists(g_0,\dots,g_k:\mathbb{R}\to\mathbb{R}),\ g_0 = f \wedge \big(\forall j \le k,\ g_j \in C^0(I)\big) \wedge \big(\forall j < k,\ g_{j+1} = (g_j)'\big)$	**Strong C^k**	**RCA$_0$**
Def 5	**Def:Diff** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(x:\mathbb{R}),\ \mathsf{Diff}(f,x) :\iff \lim_{h\to0}\frac{f(x+h)-f(x)}{h}$ exists; $\quad f'(x) :=$ that limit	**Diff**	**RCA$_0$**
Def 6	**Def:Area** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,x:\mathbb{R}),\ A^x_a :=$ $\int_a^x f(t)\,dt$ with $A^x_x = 0,\ A^{x+\mathrm{d}x}_x = -A^x_{x+\mathrm{d}x}\ (\mathrm{d}x<0)$	**Area**	**RCA$_0$**
Def 7	**Def:chord** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a,b:\mathbb{R}),\ a\neq b \to$ $\mathsf{ChordSlope}(f,a,b) := \frac{f(b)-f(a)}{b-a}$	**Chord**	**RCA$_0$**
Def 8	**Def:tangent** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{Diff}(f,c) \to$ $\mathsf{TangentSlope}(f,c) := f'(c)$	**Tangent**	**RCA$_0$**
Def 9	**Def:LocExt** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(c:\mathbb{R}),\ \mathsf{LocalExt}(f,c) :\iff \exists(\varepsilon>0),\ \big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\le f(c)\big) \vee \big(\forall(x\in(c-\varepsilon,c+\varepsilon)),\ f(x)\ge f(c)\big)$	**LocExt**	**RCA$_0$**
Def 10	**Def:Increasing** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{Increasing}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)\le f(y)$	**Incr**	**RCA$_0$**
Def 11	**Def:StrictInc** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictInc}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)<f(y)$	**SIncr**	**RCA$_0$**
Def 12	**Def:StrictDec** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictDec}(f) :\iff \forall(x,y:\mathbb{R}),\ x<y\to f(x)>f(y)$	**SDecr**	**RCA$_0$**
Def 13	**Def:StrictMono** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \mathsf{StrictMono}(f) :\iff \mathsf{StrictInc}(f)\vee\mathsf{StrictDec}(f)$	**SMono**	**RCA$_0$**
Def 14	**Def:ln** : $\forall(x:\mathbb{R}_{>0}),\ \ln(x) := \int_1^x \frac{1}{t}\,dt$	**ln**	**RCA$_0$**
Def 15	**Def:exp** : $\exp := \ln^{-1} : \mathbb{R}\to\mathbb{R}_{>0}$	**exp**	**RCA$_0$**

10. Statements

Statements and their strengths. Proofs in §11. All C^k hypotheses are strong C^k-coding (§5); pointwise differentiability appears only in Hardin–Velleman.

10.1. Topological theorems

Thm 1	**EVT** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b]) \to \exists x_{\max}\in[a,b],\ \exists x_{\min}\in[a,b],\ \forall(x\in[a,b]),\ f(x_{\min})\le f(x)\le f(x_{\max})$	**EVT**	**WKL$_0$**
Thm 2	**IVT** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b]) \wedge f(a)\cdot f(b)<0 \to \exists(c\in(a,b)),\ f(c)=0$	**IVT**	**WKL$_0$**

10.2. Differentiation theorems

Thm 3	**Fermat** : $\forall(f:\mathbb{R}\to\mathbb{R}),\ \forall(a:\mathbb{R}),\ \forall(b:\mathbb{R}),\ \forall(c:\mathbb{R}),\ a<c\wedge c<b\wedge\mathsf{LocalExt}(f,c)\wedge\mathsf{Diff}(f,c)\to f'(c)=0$	**Fermat**	**RCA$_0$**
Thm 4	**Rolle** : $\forall(f:[a,b]\to\mathbb{R}),\ f\in C^0([a,b]) \wedge f\in C^1((a,b)) \wedge f(a)=f(b)\to\exists(c\in(a,b)),\ f'(c)=0$	**Rolle**	**RCA$_0$**

Thm 5	MVT : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \to \exists (c \in (a,b)),\ f'(c) = \mathsf{ChordSlope}(f,a,b)$	MVT	RCA_0
Thm 6	Const : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \wedge (\forall x \in (a,b),\ f'(x) = 0) \to (\forall x \in [a,b],\ f(x) = f(a))$	Const	RCA_0
Thm 7	Monotone : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge f \in C^1((a,b)) \to \big((\forall x \in (a,b),\ f'(x) \geq 0) \to \mathsf{Increasing}(f)\big) \wedge \big((\forall x \in (a,b),\ f'(x) > 0) \to \mathsf{StrictInc}(f)\big)$	Mono	RCA_0
Thm 8	CauchyMVT : $\forall (f,g : [a,b] \to \mathbb{R}),\ f,g \in C^0([a,b]) \wedge f,g \in C^1((a,b)) \wedge (\forall x \in (a,b),\ g'(x) \neq 0) \to \exists (c \in (a,b)),\ \frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$	Cauchy	RCA_0
Thm 9	Darboux : $\forall (f : [a,b] \to \mathbb{R}),\ \forall (\lambda : \mathbb{R}),\ \mathsf{Diff}(f,a) \wedge \mathsf{Diff}(f,b) \wedge \big(\forall x \in (a,b),\ \mathsf{Diff}(f,x)\big) \wedge f'(a) < \lambda < f'(b) \to \exists (c \in (a,b)),\ f'(c) = \lambda$	Darboux	WKL_0
Thm 10	LHopital : $\forall (f,g : \mathbb{R} \to \mathbb{R}),\ \forall (a : \mathbb{R}),\ \forall (L : \mathbb{R}),\ f(a) = g(a) = 0 \wedge \big(\forall x \text{ near } a,\ \mathsf{Diff}(f,x) \wedge \mathsf{Diff}(g,x) \wedge g'(x) \neq 0\big) \wedge \lim_{x \to a} \frac{f'(x)}{g'(x)} = L \to \lim_{x \to a} \frac{f(x)}{g(x)} = L$	L'Hôpital	RCA_0
Thm 11	Taylor : $\forall (f : \mathbb{R} \to \mathbb{R}),\ \forall (a,x : \mathbb{R}),\ \forall (n : \mathbb{N}),\ a \neq x \wedge C^{n+1}_{\mathbf{strong}}([\min(a,x), \max(a,x)])(f) \to \exists (c \in (\min(a,x), \max(a,x))),\ f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	Taylor	RCA_0
Thm 12	Convexity : $\forall (f : [a,b] \to \mathbb{R}),\ C^2_{\mathbf{strong}}([a,b])(f) \wedge (\forall x \in (a,b),\ f''(x) \geq 0) \to \forall (x, x_0 \in [a,b]),\ f(x) \geq f(x_0) + f'(x_0)(x - x_0)$	Convex	RCA_0
Thm 13	Inverse : $\forall (f : I \to \mathbb{R}),\ \forall (x_0 \in I^\circ),\ C^1_{\mathbf{strong}}(I)(f) \wedge (\forall x \in I,\ f'(x) \neq 0) \wedge \mathsf{StrictMono}(f) \to \mathsf{Diff}(f^{-1}, f(x_0)) \wedge (f^{-1})'(f(x_0)) = 1/f'(x_0)$	Inv	RCA_0

10.3. Integration theorems

Lem 1	BarrowArea : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \wedge \mathsf{Increasing}(f) \to \forall (x_0 \in [a,b]),\ \forall (\mathrm{d}x \neq 0,\ x_0 + \mathrm{d}x \in [a,b]),\ f(x_0)\,\mathrm{d}x \leq A_{x_0}^{x_0+\mathrm{d}x} \leq f(x_0+\mathrm{d}x)\,\mathrm{d}x$	BarrowArea	RCA_0
Lem 2	MVTInt : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \to \forall (x_0, \mathrm{d}x \neq 0),\ x_0 + \mathrm{d}x \in [a,b] \to \exists (\theta \text{ between } x_0, x_0 + \mathrm{d}x),\ A_{x_0}^{x_0+\mathrm{d}x} = f(\theta)\,\mathrm{d}x$	MVTInt	RCA_0
Thm 14	FFTC : $\forall (f : [a,b] \to \mathbb{R}),\ f \in C^0([a,b]) \to \forall (x \in (a,b)),\ \frac{d}{dx} A_a^x = f(x)$	FFTC	RCA_0
Thm 15	SecondFTC : $\forall (F : [a,b] \to \mathbb{R}),\ F \in C^1([a,b]) \to \int_a^b F'(t)\,dt = F(b) - F(a)$	SecondFTC	RCA_0
Thm 16	IBP : $\forall (f,g \in C^1([a,b])),\ \int_a^b f\,g' = [fg]_a^b - \int_a^b f'\,g$	IBP	RCA_0
Thm 17	Subst : $\forall (g \in C^1([a,b])),\ \forall (f \in C^0(g([a,b]))),\ \int_{g(a)}^{g(b)} f(u)\,du = \int_a^b f(g(t))\,g'(t)\,dt$	Subst	RCA_0
Thm 18	WeightedMVT : $\forall (f \in C^0([a,b])),\ \forall (w \geq 0 \text{ Riemann integrable}),\ \int_a^b w > 0 \to \exists (c \in [a,b]),\ \int_a^b f\,w = f(c) \int_a^b w$	WMVT	WKL_0

10.4. The logarithm and the exponential

Lem 3	PosClosedMul : $\forall(a,b:\mathbb{R}),\ 0<a\wedge 0<b\to 0<a\cdot b$	Pos Closed	RCA_0
Lem 4	LogProperty : $\forall(a,b:\mathbb{R}_{>0}),\ \ln(a)+\ln(b)=\ln(a\cdot b)$	LogAdd	RCA_0
Lem 5	ExpProperty : $\forall(u,v:\mathbb{R}),\ \exp(u)\cdot\exp(v)=\exp(u+v)$	ExpAdd	RCA_0
Thm 19	ImageExpLn : $\exp(\mathbb{R})=\mathbb{R}_{>0}\wedge\ln(\mathbb{R}_{>0})=\mathbb{R}$	Image	RCA_0

10.5. Illustrative theorems

Theorem	Strength	Choice
IVT	WKL_0	none
MVT	RCA_0	none
Hahn–Banach	WKL_0	none
Weierstrass	WKL_0	none
Peano	WKL_0	none
separation	ACA_0	none
Banach–Alaoglu	ZFC	full AC
Tychonoff	ZFC	full AC

Remark 2 (EVT tag).

EVT $\equiv$ ***WKL***$_0$ *over* ***RCA***$_0$ [4]. *Entries whose printed proofs invoke EVT transitively: Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Darboux, WeightedMVT; Part 1 proofs are at most* ***WKL***$_0$ *as written. Rolle and MVT are* ***RCA***$_0$ *via Hardin–Velleman* [3]*; the descendants inherit* ***RCA***$_0$ *under strong-*C^k*-coding. Darboux and WeightedMVT are* ***WKL***$_0$ *as stated. Fermat, FFTC, SecondFTC, IBP, Subst, BarrowArea, MVTInt, Inverse, and the* ln/exp *chain are* ***RCA***$_0$ *(direct;* [3, 6] *for Inverse and MVTInt). IVT is* ***WKL***$_0$*-equivalent for the* sup *form; the neighbourhood argument below uses only continuity.*

11. Proofs

Starred subsections, one per proof, in statement order. Each proof is a four-column dependency table: first row Hyp, last row $\to$ I$(1,\ldots,N-1)$. Justifications use only Hyp, def, intro, Ax:, Pos:, Def:, Lem:, Thm:, Eq$(\ldots)$, or a logical rule ($\to$ I, $\wedge$E, $\vee$E, $\forall$I, $\forall$E, $\exists$I, $\exists$E, $\neg$I, $\neg$E, $\bot$I, $\bot$E, RAA, DNE, LEM). No external citation appears inside a proof table.

Proof of Theorem 1 (EVT)

1	$f\in C^0([a,b])$	hypothesis	Hyp
2	$\exists L,U,\ \forall x,\ L\le f(x)\le U$	bdd	Pos:14(1)
3	$\exists x_{\max},\ \forall x,\ f(x)\le f(x_{\max})$	max	Pos:15(1)
4	$\exists x_{\min},\ \forall x,\ f(x_{\min})\le f(x)$	min	Pos:15(1)
5	$\forall x,\ f(x_{\min})\le f(x)\le f(x_{\max})$	combine	Eq(3, 4)
6	$\top$	close	$\to$ I$(1,\ldots,5)$

Proof of Theorem 2 (IVT)

1	$f \in C^0([a,\,b]) \wedge f(a)f(b) < 0$	**hypothesis**	**Hyp**
2	$f(a) < 0 < f(b)$	**sign**	**Eq**(1)
3	$S := \{x \in [a,b] : f(x) < 0\}$ is $\mathbf{\Sigma}^0_1$	**def**	**Eq**(2)
4	$P := \{x \in [a,b] : -f(x) \le 0\}$ is $\mathbf{\Pi}^0_1$	**def**	**Eq**(2)
5	$c := \sup S = -\inf P$	**sup**	Pos:16(4)
6	$\forall n\, \exists x \in S,\ c - \frac{1}{n+1} < x \le c$	**sup**	**Eq**(5)
7	$\exists (x_n) \subset S,\ x_n \to c$ (codes of reals)	$\mathbf{\Sigma}^0_1$-choice	Pos:17(6)
8	$f(x_n) \to f(c)$	**seq**	Pos:18(1, 7)
9	$f(c) \le 0$	**lim**	**Eq**(7, 8)
10	$f(c) < 0 \to \exists \delta,\ \forall u \in (c-\delta,\, c+\delta),\ f(u) < 0$	**cont**	Def:3(1)
11	$f(c) < 0 \to (c + \delta/2 \in S \wedge c < c + \delta/2)$	**shift**	**Eq**(3, 10)
12	$f(c) < 0 \to \bot$	**contra**	**Eq**(5, 11)
13	$f(c) \ge 0$	**RAA**	**Eq**(12)
14	$f(c) = 0$	**antisym**	**Eq**(9, 13)
15	$\top$	**close**	$\to$ **I**(1, . . . , 14)

Coding note (IVT). In row 3, $S = \{x \in [a,b] : f(x) < 0\}$ is read as a $\mathbf{\Sigma}^0_1$ set of *codes* for reals, so **Pos:$\mathbf{\Sigma}^0_1$Choice** (row 7) applies at the level of codes. In row 4, $P = \{x \in [a,b] : -f(x) \le 0\}$ is the $\mathbf{\Pi}^0_1$ complement of S in $[a,b]$; **Pos:$\mathbf{\Pi}^0_1$Sup** applied to P yields $\inf P$, and $\sup S = -\inf P$ (cf. Simpson [4, §IV.2]).

Proof of Theorem 3 (Fermat)

1	local max, **Diff**$(f,\, c)$	**hypothesis**	**Hyp**
2	$\forall h > 0$ small, $(f(c+h) - f(c))/h \le 0$	**locmax**	Def:9(1)
3	$f'(c) \le 0$	**osl**	Def:5(2)
4	$\forall h < 0$ small, $(f(c+h) - f(c))/h \ge 0$	**locmax**	Def:9(1)
5	$f'(c) \ge 0$	**osl**	Def:5(4)
6	$f'(c) = 0$	**antisym**	Ax:11(3, 5)
7	$\top$	**close**	$\to$ **I**(1, . . . , 6)

Proof of Theorem 4 (Rolle)

1	$f \in C^0 \wedge f \in C^1 \wedge f(a) = f(b)$	**hypothesis**	**Hyp**
2	$\exists c,\ f'(c) = 0$	**HVRolle**	Pos:19(1)
3	$\top$	**close**	$\to$ **I**(1, 2)

Proof of Theorem 5 (MVT)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$L :=$ chord, $g := f - L$	def	Def:7(1)
3	$g \in C^0 \wedge g \in C^1 \wedge g(a) = g(b) = 0$	construct	Eq(2)
4	$\exists c,\ g'(c) = 0$	HVRolle	Pos:19(3)
5	$g'(c) = f'(c) -$ ChordSlope	diff	Eq(2)
6	$f'(c) =$ ChordSlope	combine	Eq(4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 6 (Const)

1	$f \in C^0 \wedge f \in C^1 \wedge f' \equiv 0$	hypothesis	Hyp
2	$\forall x,\ \exists c,\ f(x) - f(a) = f'(c)(x - a)$	MVT	Thm:5(1)
3	$f'(c)(x - a) = 0$	combine	Eq(1, 2)
4	$f(x) = f(a)$	combine	Eq(2, 3)
5	$\top$	close	$\to$ I(1, ..., 4)

Proof of Theorem 7 (Monotone)

1	$f \in C^0 \wedge f \in C^1$	hypothesis	Hyp
2	$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$	MVT	Thm:5(1)
3	$f' \geq 0 \to f(x_2) \geq f(x_1)$	sign	Eq(1, 2)
4	$f' > 0 \to f(x_2) > f(x_1)$	sign	Eq(1, 2)
5	$f' \geq 0 \to$ Increasing	combine	Def:10(3)
6	$f' > 0 \to$ StrictInc	combine	Def:11(4)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 8 (CauchyMVT)

1	$f, g \in C^0 \wedge f, g \in C^1 \wedge g' \neq 0$	hypothesis	Hyp
2	$h := (f(b) - f(a))g - (g(b) - g(a))f$	def	def
3	$h \in C^0 \wedge h \in C^1 \wedge h(a) = h(b)$	construct	Eq(2)
4	$\exists c,\ h'(c) = 0$	HVRolle	Pos:19(3)
5	$(f(b) - f(a))g'(c) = (g(b) - g(a))f'(c)$	diff	Eq(2, 4)
6	$g'(c) \neq 0$	hyp	Eq(1, 4)
7	$g(b) \neq g(a)$	Rolle(g)	Thm:4(1)
8	$f'(c)/g'(c) = (f(b) - f(a))/(g(b) - g(a))$	divide	Eq(5, 6, 7)
9	$\top$	close	$\to$ I(1, ..., 8)

Proof of Theorem 9 (Darboux)

1	f' exists, $f'(a) < \lambda < f'(b)$	hypothesis	Hyp
2	$g := f - \lambda x,\ g'(a) < 0 < g'(b)$	def	Eq(1)

3	$\exists h > 0,\ g(a+h) < g(a)$	$g'(a) < 0$	Def:5(2)
4	$\exists h > 0,\ g(b-h) < g(b)$	$g'(b) > 0$	Def:5(2)
5	$\exists c \in [a, b],\ \forall x,\ g(c) \le g(x)$	EVT	Thm:1(1)
6	$c \neq a \wedge c \neq b$	combine	Eq(3, 4, 5)
7	$c \in (a, b)$	combine	Eq(5, 6)
8	$g'(c) = 0$	Fermat	Thm:3(7)
9	$f'(c) = \lambda$	def	Eq(2, 8)
10	$\top$	close	$\to$ I(1, ..., 9)

Proof of Theorem 10 (LHopital)

1	$f(a) = g(a) = 0 \wedge (\forall x \text{ near } a,\ g'(x) \neq 0) \wedge \lim_{x\to a} f'/g' = L$	hypothesis	Hyp
2	$g \neq 0$ near a	Rolle	Thm:4(1)
3	$f(x)/g(x) = f'(c_x)/g'(c_x)$	Cauchy	Thm:8(1, 2)
4	$x \to a^+ \Rightarrow c_x \to a^+$	combine	Eq(3)
5	$f'/g'(c_x) \to L$	combine	Eq(1, 4)
6	$f/g(x) \to L\ (a^+)$	combine	Eq(3, 5)
7	$f/g(x) \to L\ (a^-)$ sym.	Cauchy	Thm:8(1, 2)
8	$\lim_{x\to a} f/g = L$	combine	Eq(6, 7)
9	$\top$	close	$\to$ I(1, ..., 8)

Proof of Theorem 11 (Taylor)

1	$f \in C^{n+1}$	hypothesis	Hyp
2	$P :=$ Taylor poly	def	def
3	$M := (f(x) - P(x))/(x-a)^{n+1}$	def	Eq(1, 2)
4	$h := f - P - M(t-a)^{n+1}$	def	Eq(2, 3)
5	$h^{(j)}(a) = 0\ (j \le n)$	def	Eq(2, 4)
6	$h(x) = 0$	def	Eq(3, 4)
7	$h \in C^n \wedge h \in C^{n+1}$	smooth	Eq(1, 4)
8	$\exists c_1,\ h'(c_1) = 0$	HVRolle	Pos:19(7)
9	$h'(a) = h'(c_1) = 0$	combine	Eq(5, 8)
10	$\exists c_2,\ h''(c_2) = 0$	HVRolle	Pos:19(7)
11	(iterate $n+1$ times)	HVRolle	Pos:19(7)
12	$\exists c,\ h^{(n+1)}(c) = 0$	combine	Eq(8, ..., 11)
13	$h^{(n+1)} = f^{(n+1)} - M(n+1)!$	def	Eq(2, 4)
14	$M = f^{(n+1)}(c)/(n+1)!$	combine	Eq(12, 13)
15	$f(x) =$ Taylor sum $+ \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$	combine	Eq(2, 3, 14)
16	$\top$	close	$\to$ I(1, ..., 15)

Proof of Theorem 12 (Convexity)

1	$f \in C^2 \land f'' \geq 0$	hypothesis	Hyp
2	$g := f - f(x_0) - f'(x_0)(x - x_0)$	def	Def:8(1)
3	$g'' = f'' \geq 0$	diff	Eq(1, 2)
4	g' nondecreasing, $g'(x_0) = 0$	Monotone	Thm:7(3)
5	$g' \leq 0$ $(x < x_0)$, $g' \geq 0$ $(x > x_0)$	split	Eq(4)
6	g nonincr./nondecr.	Monotone	Thm:7(5)
7	$g(x_0) = 0$	def	Eq(2)
8	$g \geq 0$	combine	Eq(6, 7)
9	$f(x) \geq f(x_0) + f'(x_0)(x - x_0)$	combine	Eq(2, 8)
10	$\top$	close	$\to$ I(1, ..., 9)

Proof of Theorem 13 (Inverse)

1	$x_0 \in I^\circ \land C^1_{\text{strong}}(I)(f) \land (\forall x : I,\ f'(x) \neq 0) \land \text{StrictMono}(f)$	hypothesis	Hyp
2	$\text{Diff}(f^{-1}, f(x_0)) \land (f^{-1})'(f(x_0)) = 1/f'(x_0)$	Yokoyama	Pos:20(1)
3	$\top$	close	$\to$ I(1, 2)

Proof of Lemma 1 (BarrowArea)

1	$f \in C^0([a, b])$, $\text{Incr}(f)$, $x_0 < x_0 + \mathrm{d}x$	hypothesis	Hyp
2	$f \in L^1([a, b])$	cont	Pos:5(1)
3	$f(x_0) \leq f \leq f(x_0 + \mathrm{d}x)$	incr	Def:10(1)
4	inscribed $\leq A \leq$ circumscribed	mono	Pos:3(2, 3)
5	$f(x_0)\mathrm{d}x \leq A \leq f(x_0 + \mathrm{d}x)\mathrm{d}x$	rect	Eq(4)
6	$\mathrm{d}x < 0 :\ A = -A'$	orient	Def:6
7	same inequality	negate	Eq(6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Lemma 2 (MVTInt)

1	$f \in C^0$, $x_0, x_0 + \mathrm{d}x \in [a, b]$	hypothesis	Hyp
2	$G(t) := A^t_{x_0}$	area	Def:6(1)
3	$G \in C^0 \land G' = f$	FFTC	Thm:14(1)
4	$\exists \theta,\ G'(\theta) = (G(x_0 + \mathrm{d}x) - G(x_0))/\mathrm{d}x$	MVT	Thm:5(1, 3)
5	$G(x_0) = 0 \land G(x_0 + \mathrm{d}x) = A$	def	Def:6(2)
6	$A = f(\theta)\mathrm{d}x$	combine	Eq(3, 4, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Theorem 14 (FFTC)

1	$f \in C^0,\ x_0 \in (a,b),\ \varepsilon > 0$	hypothesis	Hyp
2	$f \in L^1([a,b])$	cont	Pos:5(1)
3	$f \in L^1([\min(a, x_0, x_0+h), \max(a, x_0, x_0+h)])$	restrict	Eq(2)
4	$A_a^{x_0+h} = A_a^{x_0} + A_{x_0}^{x_0+h}$	additivity	Pos:2(3)
5	$\exists m_\varepsilon, M_\varepsilon,\ \forall u \in I_h,\ m_\varepsilon \le f(u) \le M_\varepsilon,\ \lvert m_\varepsilon - f(x_0)\rvert, \lvert M_\varepsilon - f(x_0)\rvert < \varepsilon$	cont	Def:3(1)
6	$h > 0:\ m_\varepsilon h \le A \le M_\varepsilon h$	mono	Pos:3(3,5)
7	$h < 0:\ M_\varepsilon h \le A \le m_\varepsilon h$	orient	Def:6
8	$m_\varepsilon \le (A_a^{x_0+h} - A_a^{x_0})/h \le M_\varepsilon$	divide	Eq(4, 6, 7)
9	$\mathrm{d}A_a^x/\mathrm{d}x\rvert_{x_0} = f(x_0)$	squeeze	Eq(5, 8)
10	x_0 arbitrary	gen	Eq(9)
11	$\top$	close	$\to$ I(1, ..., 10)

Proof of Theorem 15 (SecondFTC)

1	$F \in C^1,\ f := F' \in C^0$	hypothesis	Hyp
2	$G(x) := \int_a^x f,\ G' = f$	FFTC	Thm:14(1)
3	$(F-G)' = 0$	diff	Eq(1, 2)
4	$F - G$ constant	Const	Thm:6(3)
5	$F(b) - F(a) = \int_a^b f$	combine	Eq(4)
6	$\top$	close	$\to$ I(1, ..., 5)

Proof of Theorem 16 (IBP)

1	$f, g \in C^1$	hypothesis	Hyp
2	$f'g,\ fg' \in C^0([a,b]) \subseteq L^1([a,b])$	cont	Pos:5(1)
3	$(fg)' = f'g + fg'$	prod	Pos:7(1)
4	$\int_a^b (fg)' = [fg]_a^b$	SecondFTC	Thm:15(2)
5	$\int (f'g + fg') = \int f'g + \int fg'$	linear	Pos:1(2,4)
6	$[fg]_a^b = \int f'g + \int fg'$	combine	Eq(4, 5)
7	$\int fg' = [fg]_a^b - \int f'g$	rearrange	Eq(6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Theorem 17 (Subst)

1	$g \in C^1,\ f \in C^0(g([a,b]))$	hypothesis	Hyp
2	$f \in L^1(g([a,b])) \wedge f(g)g' \in L^1([a,b])$	cont	Pos:5(1)
3	$F := \int_{g(a)}^{\cdot} f,\ F' = f$	FFTC	Thm:14(1)
4	$(F \circ g)' = f(g)g'$	chain	Pos:8(1,3)

5	$\int f(g)g' = F(g(b)) - F(g(a))$	SecondFTC	Thm:15(4)
6	$\int_{g(a)}^{g(b)} f = F(g(b)) - F(g(a))$	SecondFTC	Thm:15(3)
7	both $= F(g(b)) - F(g(a))$	combine	Eq(5, 6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Theorem 18 (WeightedMVT)

1	$f \in C^0,\ w \geq 0$ int., $\int w > 0$	hypothesis	Hyp
2	$f,\ fw \in L^1([a, b])$	int	Pos:5(1)
3	$m := \min f,\ M := \max f$	EVT	Thm:1(1)
4	$m \int w \leq \int fw \leq M \int w$	mono	Pos:3(2, 3)
5	$A := \int fw / \int w \in [m, M]$	divide	Eq(4)
6	$\exists c,\ f(c) = A$	IVT	Thm:2(1, 5)
7	$\int fw = f(c) \int w$	combine	Eq(5, 6)
8	$\top$	close	$\to$ I(1, ..., 7)

Proof of Lemma 3 (PosClosedMul)

1	$0 < a,\ 0 < b$	hypothesis	Hyp
2	$0 \leq a,\ 0 \leq b$	weaken	Ax:11(1)
3	$0 \leq ab$	mulMon	Ax:13(2)
4	$a \neq 0,\ b \neq 0$	combine	Ax:11(2)
5	$ab \neq 0$	combine	Ax:8(4)
6	$0 < ab$	combine	Eq(3, 5)
7	$\top$	close	$\to$ I(1, ..., 6)

Proof of Lemma 4 (LogProperty)

1	$a,\ b > 0$	hypothesis	Hyp
2	$ab > 0$	closed	Lem:3(1)
3	$(t \mapsto 1/t) \in C^0(\mathbb{R}_{>0})$	InvCont	Pos:9
4	$(t \mapsto 1/t) \in L^1([\min(1, b, ab), \max(1, b, ab)])$	ContInt	Pos:5(3)
5	$\int_1^{ab} \frac{\mathrm{d}t}{t} = \int_1^b \frac{\mathrm{d}t}{t} + \int_b^{ab} \frac{\mathrm{d}t}{t}$	additivity	Pos:2(4)
6	$\ln(ab) = \int_1^{ab} \frac{\mathrm{d}t}{t},\ \ln(b) = \int_1^b \frac{\mathrm{d}t}{t}$	def	Def:14(1, 2)
7	$\ln(ab) - \ln(b) = \int_b^{ab} \frac{\mathrm{d}t}{t}$	combine	Eq(5, 6)
8	$g := (u \mapsto bu) \in C^1([\min(1, a), \max(1, a)]),\ g' = b \neq 0$	lin	Ax:6(1, 2)
9	$\int_b^{ab} \frac{\mathrm{d}t}{t} = \int_1^a \frac{1}{bu}\, b\, \mathrm{d}u$	subst	Pos:4(3, 8)
10	$\frac{1}{bu} \cdot b = \frac{1}{u}$	field	Ax:8(1, 2)

11	$\int_1^a \frac{1}{bu}\, b\,\mathrm{d}u = \int_1^a \frac{\mathrm{d}u}{u}$	combine	Eq(10)
12	$\int_1^a \frac{\mathrm{d}u}{u} = \ln(a)$	def	Def:14(1)
13	$\ln(ab) - \ln(b) = \ln(a)$	combine	Eq(7, 9, 11, 12)
14	$\ln(a) + \ln(b) = \ln(ab)$	rearrange	Eq(13)
15	$\top$	close	$\to$ I(1, . . . , 14)

Proof of Lemma 5 (ExpProperty)

1	$u,\ v \in \mathbb{R}$	hypothesis	Hyp
2	$a := \exp u,\ \ b := \exp v$	def	def
3	$a,\ b > 0$	def	Def:15(2)
4	$\ln a = u,\ \ \ln b = v$	inv	Def:15(2, 3)
5	$ab > 0$	closed	Lem:3(3)
6	$\ln(ab) = \ln a + \ln b = u + v$	Log	Lem:4(3, 4, 5)
7	$\exp(u + v) = \exp(\ln(ab)) = ab$	inv	Def:15(2, 6)
8	$ab = \exp u \cdot \exp v$	def	Eq(2)
9	$\exp u \cdot \exp v = \exp(u + v)$	combine	Eq(7, 8)
10	$\top$	close	$\to$ I(1, . . . , 9)

Citation note (ExpProperty). Row 5 cites **Lem:PosClosedMul(3):** current-table row 3 $(a, b > 0)$ supplies the positivity input to **PosClosedMul**, yielding $ab > 0$ at row 5. Row 6 cites **Lem:LogProperty(3,4,5):** current-table row 3 $(a, b > 0)$ supplies **LogProperty**'s positivity precondition; row 4 $(\ln a = u,\ \ln b = v)$ supplies the logarithmic values; row 5 $(ab > 0)$ makes $\ln(ab)$ well-typed in **LogProperty**'s conclusion. Including row 3 is explicit rather than redundant: **LogProperty**'s hypothesis $a, b : \mathbb{R}_{>0}$ is satisfied by row 3, even though $a =_{\mathrm{def}} \exp u$ already forces $a > 0$.

Proof of Theorem 19 (ImageExpLn)

1	$\forall u,\ \ u < 0 \vee u \geq 0$	tot	Ax:11(1)
2	$\mathbb{R} = \mathbb{R}_{<0} \cup \mathbb{R}_{\geq 0}$	part	Def:1(1)
3	$\exp(\mathbb{R}) = \exp(\mathbb{R}_{<0}) \cup \exp(\mathbb{R}_{\geq 0})$	ImgUnion	Pos:13(2)
4	$\ln(\exp u) = u$	inv	Def:15(1)
5	$\exp(u) < 1 \wedge \exp(u) > 0\ (u < 0)$	Sign	Pos:12(4, 3)
6	$\exp(\mathbb{R}_{<0}) =]0,\ 1[$	bij	Pos:11(5)
7	$\exp(u) \geq 1 \wedge \exp(u) > 0\ (u \geq 0)$	Sign	Pos:12(4)
8	$\exp(\mathbb{R}_{\geq 0}) = [1,\ \infty[$	bij	Pos:11(7)
9	$\exp(\mathbb{R}) =]0,\ \infty[$	combine	Eq(3, 6, 8)
10	$\exp(\mathbb{R}) = \mathbb{R}_{>0}$	def	Def:1(9)
11	$\ln(\mathbb{R}_{>0}) = \mathbb{R}$	bij	Pos:11(1)
12	$\top$	close	$\to$ I(1, . . . , 11)

12. Scripts

Each tree lists entries used by a proof, transitively closed, with strengths. `L0` **is the root.**

12.1. Foundational leaves

Axioms (all ZF).

```
ax:R-add-assoc      (ZF)
ax:R-add-comm       (ZF)
ax:R-add-zero       (ZF)
ax:R-add-inv        (ZF)
ax:R-mul-assoc      (ZF)
ax:R-mul-comm       (ZF)
ax:R-mul-one        (ZF)
ax:R-mul-inv        (ZF)
ax:R-distrib        (ZF)
ax:R-nontriv        (ZF)
ax:R-order          (ZF)
ax:R-add-monotone   (ZF)
ax:R-mul-monotone-pos (ZF)
ax:R-complete       (ZF, Cut)
```

Postulates.

```
Integral:
Pos:IntLinear       (RCA0)
Pos:IntAdd          (RCA0)
Pos:IntMono         (RCA0)
Pos:IntSubst        (RCA0)
Pos:ContInt         (RCA0)
Pos:Similarity      (RCA0)

Differential:
Pos:ProdRule        (RCA0)
Pos:ChainRule       (RCA0)
Pos:InvCont         (RCA0)
Pos:BarrowTangency (RCA0)

Log/Logic:
Pos:LnBij           (RCA0)
Pos:LnSign          (RCA0)
Pos:ImgUnion        (FOL)

External:
Pos:ContImgBdd      (RCA0)
Pos:ContImgAttain   (WKL0)
Pos:Pi01Sup         (WKL0)
Pos:Sigma01Choice   (RCA0)
Pos:SeqCont         (RCA0)
Pos:HVRolle         (RCA0)
Pos:YokoyamaInv     (RCA0)
```

12.2. Per-result trees

Lemma **PosClosedMul** (**RCA$_0$**)

```
L0 | PosClosedMul                  (RCA0)
L1 |   ax:R-order                  (ZF)
L1 |   ax:R-mul-monotone-pos        (ZF)
L1 |   ax:R-mul-inv                 (ZF)
```

Lemma **LogProperty** (**RCA$_0$**)

```
L0 | LogProperty                   (RCA0)
L1 |   PosClosedMul                (RCA0)
L1 |   Pos:IntLinear               (RCA0)
L1 |   Pos:IntAdd                  (RCA0)
L1 |   Pos:IntSubst                (RCA0)
L1 |   Def:ln                      (RCA0)
L1 |   ax:R-mul-comm               (ZF)
L1 |   ax:R-mul-inv                (ZF)
```

Lemma **ExpProperty** (**RCA$_0$**)

```
L0 | ExpProperty                   (RCA0)
L1 |   LogProperty                 (RCA0)
L1 |   PosClosedMul                (RCA0)
L1 |   Def:exp                     (RCA0)
```

Theorem **ImageExpLn** (**RCA$_0$**)

```
L0 | ImageExpLn                    (RCA0)
L1 |   ax:R-order                  (ZF)
L1 |   Pos:ImgUnion                (FOL)
L1 |   Pos:LnBij                   (RCA0)
L1 |   Pos:LnSign                  (RCA0)
L1 |   Def:exp                     (RCA0)
```

```
L1 |   Def:ln                          (RCA0)
L1 |   Def:R-pos                       (RCA0)
```

Theorem **MVT**: two proofs, two bounds

Proof A (Barrow, §3; WKL$_0$).

```
L0 | MVT (Proof A)                    (WKL0)
L1 |   EVT                            (WKL0)
L1 |   Pos:BarrowTangency             (RCA0)
L1 |   Def:chord                       (RCA0)
```

Proof B (Rolle-based, H–V; RCA$_0$).

```
L0 | MVT (Proof B)                    (RCA0)
L1 |   Pos:HVRolle                    (RCA0)
L1 |   Def:chord                       (RCA0)
```

The tag RCA$_0$ is the strength of Proof B; Proof A alone is WKL$_0$. Same split for Rolle, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity. Darboux and WeightedMVT are WKL$_0$ throughout.

Theorem **FFTC** (RCA$_0$)

```
L0 | FFTC                             (RCA0)
L1 |   BarrowArea                     (RCA0)
L2 |     Pos:IntMono                  (RCA0)
L1 |   Pos:IntAdd                     (RCA0)
L1 |   Def:Area                       (RCA0)
L1 |   Def:Ck                         (RCA0)
```

12.3. Transitive closure

```
ImageExpLn                          (RCA0)
|-- ax:R-order                      (ZF)
|-- Pos:ImgUnion                    (FOL)
|-- Pos:LnBij                       (RCA0)
|-- Pos:LnSign                      (RCA0)
|-- Def:exp                         (RCA0)
|-- Def:ln                          (RCA0)
`-- Def:R-pos                       (RCA0)

ExpProperty                         (RCA0)
|-- LogProperty                     (RCA0)
|-- PosClosedMul                    (RCA0)
`-- Def:exp                         (RCA0)

LogProperty                         (RCA0)
|-- PosClosedMul                    (RCA0)
|-- Pos:IntLinear                   (RCA0)
|-- Pos:IntAdd                      (RCA0)
|-- Pos:IntSubst                    (RCA0)
|-- Def:ln                          (RCA0)
|-- ax:R-mul-comm                   (ZF)
`-- ax:R-mul-inv                    (ZF)

PosClosedMul                        (RCA0)
|-- ax:R-order                      (ZF)
|-- ax:R-mul-monotone-pos           (ZF)
`-- ax:R-mul-inv                    (ZF)
```

12.4. Strength map

```
STRENGTH  |  ENTRIES
----------+-------------------------------------------------------------
FOL       |  Pos:ImgUnion
```

```
----------+-------------------------------------------------------------
        RCA0      |  POSTULATES: IntLinear IntAdd IntMono IntSubst ContInt
|              Similarity ProdRule ChainRule InvCont
|              BarrowTangency LnBij LnSign
|              ContImgBdd Sigma01Choice SeqCont HVRolle YokoyamaInv
|  DEFS: L1 R-pos Ck strong-Ck Diff Area Chord Tangent
|        LocExt Increasing StrictInc StrictDec StrictMono ln exp
|  TOPOLOGY:  (none)
|  DIFF: Fermat Rolle MVT Const Monotone CauchyMVT
|        LHopital Taylor Convexity Inverse
|  INTEG: BarrowArea MVTInt FFTC SecondFTC IBP Subst
|  LOG/EXP: PosClosedMul LogProperty ExpProperty ImageExpLn
----------+-------------------------------------------------------------
WKL0      |  IVT, EVT, Darboux, WeightedMVT
|  (as printed; see EVT tag remark in \S\ref{sec:core-statements})
|  Pos:ContImgAttain, Pos:Pi01Sup
|  IVT (seq), MVT (seq)
|  Hahn-Banach, Weierstrass
----------+-------------------------------------------------------------
ACA0      |  separation (closed convex)
----------+-------------------------------------------------------------
ZF        |  ax:R-add-assoc ... ax:R-complete (all 14)
----------+-------------------------------------------------------------
ZFC       |  Banach-Alaoglu, Tychonoff, well-ordering
----------+-------------------------------------------------------------
```

12.5. Dependency depth

Axioms and postulates are depth 0; definitions depth 0 as primitive notation.

Depth	Entries (selection)
0	13 axioms used in Part II (14th, Cut, unused), 20 postulates, 15 definitions
1	PosClosedMul, Fermat, BarrowArea
2	Rolle, MVT, LogProperty, ImageExpLn, MVTInt, FFTC
3	ExpProperty, Const, Monotone, SecondFTC
4	CauchyMVT, Convexity, IBP, Subst

Longest chain: $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$ **(length 3). Diff/integ chains reach depth 4.**

12.6. Strength propagation

Result	Lower	Upper	Choice
IVT, EVT	WKL_0	WKL_0	none
Fermat	RCA_0	RCA_0	none
Rolle, MVT	RCA_0	RCA_0	none
Const, Monotone	RCA_0	RCA_0	none
CauchyMVT, LHopital, Taylor	RCA_0	RCA_0	none
Convexity	RCA_0	RCA_0	none
Inverse, MVTInt	RCA_0	RCA_0	none
Darboux, WeightedMVT	WKL_0	WKL_0	none
BarrowArea, FFTC	RCA_0	RCA_0	none

Bounds coincide: every RCA_0 tag has an RCA_0 proof in §11 (direct, or Hardin–Velleman, or Yokoyama); every WKL_0 tag has the classical proof. Part 1's geometric proofs of Rolle, MVT, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity are at most WKL_0 as written; the gap is disclosed in the EVT remark and does not affect any tag. No proof of Part 2 uses the Cut axiom.

13. Conclusion

Classical analysis developed in Barrow's geometry (Part 1) is classified in reverse mathematics (Part 2):

- 14 field axioms: **ZF**, no choice.
- Postulates: RCA_0-provable or first-order logical.
- RCA_0: Rolle, MVT, Fermat, Const, Monotone, CauchyMVT, LHopital, Taylor, Convexity, Inverse, FFTC, SecondFTC, IBP, Subst, ln/exp addition and image.
- WKL_0 as proved here: IVT, EVT, Darboux, WeightedMVT.
- No Part 2 entry uses Cauchy completeness ($\mathsf{DC}_{\mathbb{R}}$-equiv) or Cut.

13.1. Wider hierarchy

Single-function MVT is RCA_0 [3]. IVT, EVT, and sequential IVT/MVT are WKL_0 [4, 5]. Hahn–Banach and Weierstrass are WKL_0 [4]. Separation is ACA_0 [15]. Banach–Alaoglu, Tychonoff, and well-ordering are **ZFC** [12, 13, 14]. Every Part 1 entry is choice-free.

13.2. Dependency analysis

The core chain has depth 3 with longest path $ax{:}R\text{-}order \to PosClosedMul \to LogProperty \to ExpProperty$; differentiation and integration chains reach depth 4.

The load-bearing nodes are **ax:R-order**, **ax:R-mul-inv**, **Def:ln**, **Def:exp**, **Pos:LnBij**, **PosClosedMul**.

The join of predecessor strengths never exceeds WKL_0 and never reaches $\mathsf{ZF} + \mathsf{DC}_{\mathbb{R}}$. Classical theorems are exactly those whose proofs need only area additivity Equation 5, rectangle comparison Equation 31, the differentiating postulates Equation 10–Equation 11, and (for the four WKL_0 entries) the standard WKL_0-strength principles. No choice principle is needed.

13.3. Formal verification

Lean 4 and Mathlib [10] contain many of these results; about 70.4% of Mathlib declarations depend syntactically on **Classical.choice**, of roughly half of which reach it only through classical logic rather than genuine choice, and 13.1% of theorems have a choice-free statement with a choice-dependent proof [11]. The classification above is strictly stronger: it certifies the choice-free proofs.

References

[1] I. Barrow. *The Geometrical Lectures of Isaac Barrow*. Translated, with notes and proofs, by J. M. Child. Open Court Series of Classics of Science and Philosophy, No. 3, Open Court Publishing Company, Chicago, 1916. https://archive.org/details/geometricallectu00barriala.

[2] J. M. López Fernández. *A commentary on Barrow's proof of the First Fundamental Theorem of Calculus*. Manuscript, 2026. https://www.academia.edu/24088978/A_Commentary_on_Barrow_s_Proof_of_the_First_Fundamental_Theorem_of_Calculus.

[3] C. S. Hardin and D. J. Velleman. The mean value theorem in second order arithmetic. *J. Symbolic Logic* 66 (2001), no. 3, 1353–1358. https://doi.org/10.2307/2694974.

[4] S. G. Simpson. *Subsystems of Second Order Arithmetic*, 2nd ed. Perspectives in Logic, Association for Symbolic Logic, Cambridge University Press, Cambridge, 2009. ISBN 978-0-521-88439-6. https://doi.org/10.1017/CBO9780511581007.

[5] S. Sanders. Reverse-engineering reverse mathematics. *Ann. Pure Appl. Logic* (2020). https://arxiv.org/abs/2001.07513.

[6] K. Yokoyama. Differential calculus in second order arithmetic. Mathematical Institute, Tohoku University, 2010. https://researchmap.jp/read0145758/misc/9319571/attachment_file.pdf.

[7] J. Propp. Real analysis in reverse. *Amer. Math. Monthly* 119 (2012), no. 5, 371–386. https://arxiv.org/abs/1204.4483.

[8] H. Teismann. Toward a more complete list of completeness axioms. *Amer. Math. Monthly* 120 (2013), no. 2, 99–114. https://doi.org/10.4169/amer.math.monthly.120.02.099.

[9] D. Bridges, H. Ishihara, M. Rathjen, and H. Schwichtenberg. *Handbook of Constructive Mathematics*. Cambridge University Press, Cambridge, 2023. https://doi.org/10.1017/9781009039888.

[10] The Lean community. *Lean 4 Theorem Prover and Mathlib*. https://leanprover-community.github.io/.

[11] L. J. Close. Where Formal Libraries Spend Their Axioms: A Cross-Foundation Measurement, and an Avoidable Classical Dependency in Lean's omega. Zenodo, 2026. https://doi.org/10.5281/zenodo.20710920.

[12] L. Alaoglu. Weak topologies of normed linear spaces. *Ann. of Math.* 41 (1940), no. 2, 252–267. https://doi.org/10.2307/1968821.

[13] J. L. Kelley. The Tychonoff product theorem implies the axiom of choice. *Fund. Math.* 37 (1950), 75–76. https://doi.org/10.4064/fm-37-1-75-76.

[14] T. J. Jech. *The Axiom of Choice*. North-Holland, Amsterdam, 1973. https://www.sciencedirect.com/book/9780720425055/the-axiom-of-choice.

[15] A. J. Humphreys and S. G. Simpson. Separation and weak König's lemma. *Trans. Amer. Math. Soc.* 368 (2016), no. 9, 6141–6162. https://doi.org/10.1090/tran/6563.

[16] R. M. Solovay. A model of set theory in which every set of reals is Lebesgue measurable. *Ann. of Math.* 92 (1970), 1–56. https://doi.org/10.2307/1970696.